



GENERAL PERMIT 12 (GP12)

Oil and Gas - WELL PRODUCTION FACILITIES

Issuance 1

Final Approval

Date Issued: May 28, 2026

Bradley Eades, PE

Oil and Gas Permitting Section Supervisor

Permit History

Issuance	Date	Description
Issuance 1	This Issuance	Initial issuance



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Table of Contents

I.	General Permit Applicability.....	4
II.	General Operating Conditions	10
III.	Annual Emission Limitation.....	11
IV.	Annual Process Limitations.....	14
V.	State and Federal Regulatory Requirements	19
VI.	Control, Operating, and Maintenance Requirements.....	30
VII.	Initial and Periodic Compliance Test Requirements.....	45
VIII.	Recordkeeping and Reporting Requirements	59
IX.	Alternative Operating Scenarios (AOS)	65
X.	General Permit Registration Approval Process and Expiration.....	72
XI.	General Permits Terms and Administration.....	73
XII.	Additional Requirements	78
Attachment A: Alternative Operating Scenarios - Natural gas-fired reciprocating internal combustion engines		81
Attachment B: Alternative Operating Scenarios - Diesel fuel-fired reciprocating internal combustion engines		98

Note: See PS Memos and the Emission Calculation Manual for further information on definitions and emission calculations.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

This general permit is granted by the Colorado Department of Public Health and Environment, Air Pollution Control Division (Division) subject to all rules and regulations of the Colorado Air Quality Control Commission (Commission) and the Colorado Air Pollution Prevention and Control Act (C.R.S. 25-7-101 et seq), to those general terms and conditions included in this general permit, and the following specific terms and conditions:



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

I. General Permit Applicability

- I.A This general permit may be used only for oil and gas well production facilities as defined in Regulation Number 7, Part B, Section I.B.33 and Section II.A.46 (Adopted: 04/18/2025, Effective: 06/14/2025). Equipment covered by this general permit is limited to the following sources:
- I.A.1 Natural gas-fired reciprocating internal combustion engines.
 - I.A.2 Diesel fuel-fired reciprocating internal combustion engines.
 - I.A.3 Storage tanks used to store condensate, crude oil, intermediate hydrocarbon liquids, or produced water:
 - I.A.3.a Condensate storage tanks with a design capacity less than or equal to 10,000 barrels per storage vessel, used for the storage of condensate.
 - I.A.3.b Crude oil storage tanks with a design capacity less than or equal to 10,000 barrels per storage vessel, used for the storage of crude oil.
 - I.A.3.c Intermediate hydrocarbon liquid storage tanks with a design capacity less than or equal to 10,000 barrels per storage vessel, used for storage of intermediate hydrocarbon liquid.
 - I.A.3.d Produced water storage tanks used for the storage of produced water.
 - I.A.4 Hydrocarbon liquid loading operations including condensate loading, crude oil, and intermediate hydrocarbon liquid loading.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- I.A.5 Gas venting from separators.
- I.A.6 Fugitive component leak emissions.
- I.A.7 Combustion devices, vapor recovery units, catalysts, or other Division-approved air pollution control equipment used to reduce emissions (from the emitting activities registered under this general permit) below the limits specified in Section III below.
- I.A.8 Routine or predictable emissions: The following emitting activities may register for this general permit:
 - I.A.8.a Fixed roof storage tank-related routine or predictable gas venting emissions events that are not separately permitted:
 - I.A.8.a.(i) Releasing emissions from a fixed-roof storage tank through a storage tank thief hatch, blowdown valve, venting point, or other access point for any routine or predictable activity, including for well level measurement, sampling, truck loadout, well unloading, equipment maintenance, repair, decommissioning, or other routine or predictable activity; and
 - I.A.8.a.(ii) Tank hot oil treatments or other tank cleaning activities.
 - I.A.8.b Well-related routine or predictable gas venting emissions events that are not separately permitted, including but not limited to:
 - I.A.8.b.(i) Wells liquids unloading;



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- I.A.8.b.(ii) Downhole well maintenance, including well workovers, recompletion, restimulation, reperforating, paraffin/scale removal, swabbing, and installation of artificial lift;
- I.A.8.b.(iii) Well equipment maintenance at the surface;
- I.A.8.b.(iv) Well plugging;
- I.A.8.b.(v) Bradenhead gas venting.
- I.A.8.b.(vi) Casinghead gas venting; and
- I.A.8.b.(vii) Other well-related routine or predictable gas venting emissions activities.
- I.A.8.c Pigging-related routine or predictable gas venting emissions that are not separately permitted, including, but not limited to:
 - I.A.8.c.(i) Pig launchers; and
 - I.A.8.c.(ii) Pig receivers.
- I.A.8.d Routine or predictable blowdowns of facility equipment or piping that are not separately permitted, including but not limited to:
 - I.A.8.d.(i) Pipeline venting within the facility boundary when such venting is routine or predictable.
 - I.A.8.d.(ii) Compressor and turbine blowdowns.
 - I.A.8.d.(iii) Scrubber and strainer blowdowns.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- I.A.8.d.(iv) Blowdowns resulting from facility-wide (plant) and/or equipment shutdowns.
- I.A.8.d.(v) Blowdown of all other equipment (including pipelines, compressor case or cylinders, manifolds, suction bottles, discharge bottles, and vessels); and
- I.A.8.d.(vi) Desiccant dehydration unit blowdowns.
 - I.A.8.d.vi.(A) Emissions resulting from blowdowns from facility equipment or piping where the physical volume between isolation valves is less than 50 cubic feet and that are not required to be tracked and reported for the purposes of Regulation Number 7, Part B, Section V.C.2.d. and not required to be reported or permitted with this category.
- I.A.8.e This category DOES NOT include:
 - I.A.8.e.(i) Emissions at a storage tank that are not listed in Section I.A.8 of this general permit.
 - I.A.8.e.(ii) Venting emissions from a storage tank that are prohibited by Regulation Number 7, Part B, Section II.C.2.a.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- I.B The owner or operator of any oil and gas well production facility that can comply with all of the conditions described in this general permit and meet all of the requirements in this Section I may register for this general permit.
- I.C This general permit applies only to the well production facility and the equipment described in Condition I.A. for which the operator has registered under this general permit in the application materials submitted to the Division.
- I.D This general permit applies only to the equipment as described in Condition I.A above having uncontrolled actual emissions of Sulfur Oxides (SO_x), Particulate Matter (PM), Particulate Matter 2.5 (PM_{2.5}), and Particulate Matter 10 (PM₁₀) less than the Air Pollutant Emission Notice (APEN) reporting thresholds for each pollutant in Regulation Number 3, Part A, Section II.D.1.a.
- I.E This general permit applies only to the equipment described in Condition I.A located at a true minor source or synthetic minor source for Operating Permit (OP), New Source Review (NSR), Prevention of Significant Deterioration (PSD) and Maximum Achievable Control Technology (MACT) programs, except as provided for in Condition I.F below.
- I.F Stationary Sources that a) become subject to Title V Operating Permit requirements or become classified as an existing major stationary source due to a reclassification of an ozone non-attainment area, and b) for which a complete GP12 registration request was deemed complete by the Division prior to the effective date of a new reclassification of an ozone non-attainment area or prior to the effective date of a new ozone non-attainment area, may continue to operate under this general permit. A new reclassification is defined as any ozone non-attainment reclassification that happens after the effective date of GP12 registration as described in Condition X.A. For the purposes of this general permit, ozone non-attainment area must mean any



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

ozone non-attainment area defined per the most current version of Regulation Number 7, Part A, Section II.A. or Regulation Number 26, Part A, Section I.A.

- I.G The owner or operator must use state default emissions factors to register Hydrocarbon Liquids Loadout Operations under this general permit. The owner or operator must use state default emission factors approved in PS-Memo 14-02 issued by the Division. The owner or operator must use the state default emissions factors to show compliance with annual emission limitations established in this general permit.
- I.H Excluded Sources: The following sources may not register for this general permit:
- I.H.1 A natural gas processing plant.
 - I.H.2 A natural gas compressor station.
 - I.H.3 A pigging facility.
 - I.H.4 A class II disposal well facility.
 - I.H.5 An oil stabilization facility.
 - I.H.6 A petroleum refinery.
 - I.H.7 A portable source.
 - I.H.8 Equipment subject to a New Source Performance Standard (NSPS), other than NSPS Subparts 0000, 0000a, 0000b, JJJJ, and IIII.
 - I.H.9 Other equipment at the same stationary source not described in Condition I.A above must be permitted separately as required by Regulation Number 3, Part B.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

I.H.10 This general permit may not be used for new major stationary source well production facilities or modifications at existing major stationary source well production facilities that are subject to Regulation Number 3, Part D Prevention of Significant Deterioration (PSD) or Nonattainment New Source Review (NA NSR).

II. General Operating Conditions

- II.A The owner or operator must notify the Division no later than fifteen (15) days after commencement of operation of an emissions unit under this general permit by submitting a Notice of Startup (NOS) to the Division. The NOS must be submitted using the Division's online submission tool on the Division's self-certification webpage. Individual NOSs must be submitted per emissions unit. Failure to notify the Division of startup of the permitted source is a violation of Regulation Number 3, Part B, Section III.G.1. and can result in the revocation of the permit.
- II.B The general permit registration expires if the owner or operator of the sources for which this general permit was issued: (i) does not commence construction, modification or operation of these sources within 18 months after either the date of completeness of the general permit registration or the date on which such construction or activity was scheduled to commence as set forth in the permit application associated with this general permit; (ii) discontinues construction for a period of eighteen months or more; (iii) does not complete construction within a reasonable time of the estimated completion date. The Division may grant extensions of the deadline. (Regulation Number 3, Part B, Section III.F.4.)
- II.C The operator must complete all initial compliance testing and sampling as required in this general permit.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- II.D Air pollution control equipment may be used to comply with the emission limits and performance standards required in this general permit. (Regulation Number 7, Part B, Section I.B.2.)
- II.E This general permit may not be used to grant an overall control efficiency from enclosed combustion devices over 95% for Volatile Organic Compounds (VOC).
- II.F The owner or operator must construct and operate the facility in accordance with the general permit registration approval letter, or if an approval letter has not been issued yet, as certified by the owner or operator and/or required in the approved GP12 modeling workbook, the approved APCD Form 114, or the approved ambient air impact analysis (whichever is applicable). This includes, but is not limited to, stack heights, stack diameters, source's location relative to the ambient air boundary, emission rates, and preclusion of public access to the site.

III. Annual Emission Limitation

III.A Equipment Specific Emission Limitations

- III.A.1 The maximum allowable VOC, NO_x, and CO emissions from each emission point registered under this general permit must not exceed the requested annual emission limitations (tons per year) as listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the Air Pollutant Emission Notices (APEN) included with the general permit registration. (Regulation Number 3, Part B, Sections II.A.4. and II.A.7.)
- III.A.2 The owner or operator must operate and maintain the equipment registered under the general permit with the air pollution control equipment as listed in the general permit registration together with the process limitations in



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Section IV to reduce emissions to less than or equal to the emissions limits established in this general permit. (Regulation Number 3, Part B, Sections II.A.4. and II.A.7.)

III.B Facility-Wide Emission Limitations

III.B.1 The maximum allowable emissions of hazardous air pollutants (HAPs) from all permitted emission points at the stationary source must not exceed the following limitations. (Regulation Number 3, Part B, Sections II.A.4. and II.A.7.)

III.B.1.a Each Individual HAP: 8.0 tons per year

III.B.1.b Combined total of all HAPs: 20.0 tons per year

III.C For sources located at true minor facilities, compliance with the annual emission limitations in the Equipment Specific Emission Limitations and Facility-Wide Emission Limitations conditions above for both criteria and hazardous air pollutants must be determined on an annual calendar-year basis, unless required elsewhere in this general permit. The permit holder must keep a compliance record on-site or at a local field office for Division review.

III.D For sources located at synthetic minor or major source subject to Condition I.F, compliance with the Equipment Specific Emission Limitations and Facility-Wide Emission Limitations for both criteria and hazardous air pollutants must be determined on a rolling twelve (12) month total basis. By the end of each month, a new twelve-month total is calculated based on the previous twelve months' data. During the first year of operation, compliance must be based upon cumulative emissions from all months of operation. The permit holder must calculate actual



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

emissions each month and keep a compliance record on-site or at a local field office for Division review.

- III.E The owner or operator of Storage Tanks, Gas Venting from Separators, and Routine or Predictable Emissions must use the most recently developed emission factors as required by the Initial and Periodic Compliance Test Requirements section of this general permit to calculate actual emissions and to show compliance with the limits in this general permit. If a source is not subject to initial or periodic testing or prior to the initial compliance deadlines outlined in the Initial and Periodic Compliance Test Requirement section of this general permit, the owner or operator must use the emission factors listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN(s) included with the general permit registration to calculate actual emissions and to show compliance with the limits in this permit.
- III.F The owner or operator of Natural Gas-Fired Reciprocating Internal Combustion Engines must calculate actual emissions from each engine using the following parameters and equations. For the purposes of this condition, placed in service means the bringing of an engine on-site for use. The placed in service date is the date the engine begins to operate. Replacement under an authorized alternative operating scenario is not considered placed in service.
- III.F.1 The emission factors as listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN(s) included with the general permit registration;
- III.F.2 actual monthly fuel consumption as determined by Condition IV.A.6, and



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

III.F.3 for engines placed in service on or after the issuance date of Issuance 1 of this permit, the owner or operator must use the fuel gas heat content obtained from the most recent site-specific analysis as required by Condition VII.E.1. For engines placed in service before the issuance date of Issuance 1 of this permit, the owner or operator must use the fuel gas heat content as listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN(s) included with the general permit registration.

$$\text{Emissions} \left(\frac{\text{ton}}{\text{month}} \right) = \frac{\text{Emission factor} \left(\frac{\text{lb}}{\text{MMBTU}} \right) \times \text{Fuel Consumption}_{\text{engine}} \left(\frac{\text{MMSCF}}{\text{month}} \right) \times \text{Fuel Gas Heat Content} \left(\frac{\text{BTU}}{\text{SCF}} \right)}{\text{Unit Conversion} \left(\frac{2000\text{lb}}{\text{ton}} \right)}$$

III.G The owner or operator must calculate Fugitive Component Leak actual emissions based on the most recent component counts for the facility and with the most recent extended gas and liquids analyses, as required in the Initial and Periodic Compliance Test Requirements section of this permit. Prior to the initial compliance deadlines outlined in the Initial and Periodic Compliance Test Requirement section of this general permit, the owner or operator must use the component counts and the gas and liquids compositions listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN(s) included with the general permit registration to calculate actual emissions and to show compliance with the limits in this general permit.

IV. Annual Process Limitations

IV.A The maximum allowable process rates of the parameters listed below from each emission point registered under this general permit must not exceed the requested annual process limitation as listed on the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

the APEN included with the general permit registration. The owner or operator must track and maintain monthly process records of the parameters listed below for each emission point registered under this general permit. The owner or operator must use monthly throughput records to demonstrate compliance with the process limits established in this condition and to demonstrate compliance with emission limits established in Conditions III.A and III.B for each emission point registered under this general permit. (Regulation Number 3, Part B, Sections II.A.4. and II.A.7.)

IV.A.1 Storage Tanks

IV.A.1.a The owner or operator must maintain monthly records of condensate, crude oil, produced water, and intermediate hydrocarbon liquid throughput.

IV.A.1.b Whenever the owner or operator collects waste gas vapors from this source through a Vapor Recovery Unit (VRU) to a process and during any VRU downtime, the owner or operator must track and maintain condensate, crude oil, produced water, and intermediate hydrocarbon liquid throughput.

IV.A.2 Hydrocarbon liquid loading operations

IV.A.2.a The owner or operator must maintain monthly records of condensate and crude oil, and intermediate hydrocarbon liquid throughput.

IV.A.2.b Whenever the owner or operator collects waste gas vapors from this source through a Vapor Recovery Unit (VRU) to a process and during any VRU downtime, the owner or operator must track and



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

maintain crude oil, condensate, and intermediate hydrocarbon liquid throughput.

IV.A.3 Gas Venting from Separators

IV.A.3.a The owner or operator must continuously track and record the volumetric flow rate of gas emitted from the separator(s) to a combustion device using a continuous operational flow meter. The owner or operator must maintain monthly records of the volumetric flow rate of gas emitted from the separator(s) to a combustion device using a flow meter. For this requirement, records do not include gas delivered to a gas gathering line. The owner or operator must follow manufacturer recommendations for the operation, maintenance, and calibration of the flow meter.

IV.A.4 Routine or Predictable Emissions

IV.A.4.a When using a volumetric process:

IV.A.4.a.(i) The owner or operator must continuously monitor and record the volumetric flow rate of gas emitted from routine or predictable activities using a continuous operational flow meter. The owner or operator must maintain monthly records of the volumetric flow rate of gas emitted from routine or predictable activities. For this requirement, records do not include gas delivered to a gas gathering line or process. The owner or operator must follow manufacturer



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

recommendations for the operation, maintenance, and calibration of the flow meter. Or;

IV.A.4.a.(ii) The owner or operator must maintain monthly records of parametric data necessary to complete calculations of emitted volume of pollutants.

IV.A.4.b When using hours of operation: The owner or operator must maintain monthly records of the hours of operation of each activity permitted under this general permit.

IV.A.4.c When using events-based processes: The owner or operator must maintain monthly records of the number of events from each activity permitted under this general permit.

IV.A.5 Combustion Devices - Pilot light gas

IV.A.5.a The owner or operator must maintain monthly records of hours of operation of all pilot lights from all combustion devices covered by this general permit or assume the pilot light operates 8760 hours per year. The owner or operator must use the monthly hours of operation of pilot lights and the pilot light emission rates in standard cubic feet per hour (scf/hr) provided by the enclosed combustion device manufacturer or estimated through engineering calculations to calculate monthly pilot light gas throughput.

IV.A.6 Natural gas-fired reciprocating internal combustion engines - the owner or operator must use one of the following compliance methods. For the purposes of this condition, placed in service means the bringing of an



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

engine on-site for use. The placed in service date is the date the engine begins to operate. Replacement under an authorized alternative operating scenario is not considered placed in service.

IV.A.6.a Engines placed in service on or after the issuance date of Issuance 1 of this permit: The owner or operator must maintain monthly records of natural gas fuel consumption. Fuel consumption must be measured and recorded using a continuous fuel flow meter at the inlet to the engine. The owner or operator must continuously monitor and record the volumetric flow rate of fuel. The owner or operator must follow manufacturer recommendations for the operation, maintenance, and calibration of the flow meter.

IV.A.6.b Engines placed in service before the issuance date of Issuance 1 of this permit: The owner or operator must maintain monthly records of hours of operation. The actual monthly process rate must be calculated using the following equation:

$$\text{Fuel Consumption}_{\text{engine}} \left(\frac{\text{MMSCF}}{\text{month}} \right) = \frac{\text{Hours of operation} \left(\frac{\text{hrs}}{\text{month}} \right) \times \text{BSFC} \left(\frac{\text{BTU}}{\text{hp hr}} \right) \times \text{Site rated HP (hp)}}{\text{Fuel Gas Heat Content} \left(\frac{\text{BTU}}{\text{SCF}} \right) \times 10^6 \left(\frac{\text{SCF}}{\text{MMSCF}} \right)}$$

IV.A.6.b.(i) BSFC is the Break Specific Fuel Consumption at 100% load as listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN(s) included with the general permit registration.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

IV.A.6.b.(ii) The owner or operator must use the fuel gas heat content as listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN(s) included with the general permit registration.

IV.A.7 Diesel fuel-fired reciprocating internal combustion engines

IV.A.7.a The owner or operator must maintain monthly records of diesel fuel consumption.

IV.B For sources located at true minor facilities, compliance with the annual process limits must be determined on an annual calendar-year basis, unless required elsewhere in this general permit. The permit holder must keep a compliance record on-site or at a local field office for division review.

IV.C For sources located at synthetic minor or major stationary source subject to Condition I.F, compliance with process limits must be determined on a rolling twelve (12) month total basis. By the end of each month, a new twelve-month total is calculated based on the previous twelve months' data. During the first year of operation, compliance must be based upon cumulative emissions from all months of operation. The permit holder must calculate actual emissions each month and keep a compliance record on-site or at a local field office for Division review.

V. State and Federal Regulatory Requirements

V.A This stationary source is subject to the odor requirements of Regulation Number 2.

V.B The permit number (GP12) must be marked at the entrance to the facility or in another easily identifiable location within the facility. Except for Routine or



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Predictable Emissions and Fugitives Component Leak Emissions, the AIRS number (e.g. 123/1234/001) must be marked on equipment for ease of identification. (Regulation Number 3, Part B, Section III.E.)

- V.C Visible emissions shall not exceed twenty percent (20%) opacity during normal operation of any source registered to operate under this general permit. During periods of startup, process modification, or adjustment of control equipment visible emissions shall not exceed 30% opacity for more than six minutes in any sixty consecutive minutes. (Regulation Number 1, Sections II.A.1. & II.A.4.)
- V.D Owners or operators of new or modified sources located in Disproportionately Impacted Communities must submit a notification at the time of registration for a general permit or permits issued pursuant to Regulation Number 3, Part B, Section III.I. indicating the source will comply with the monitoring requirements outlined in Regulation Number 3, Part B, Section III.J., as applicable, by either:
- V.D.1 Complying with source-specific monitoring pursuant to Section III.J.2. of Part B;
 - V.D.2 Paying community monitoring fees pursuant to Section III.J.3. of Part B; or
 - V.D.3 Complying with well production facility monitoring pursuant to Section III.J.4. of Part B.
- Owners or operators must comply with the monitoring methodology selected on the general permit registration form.
- V.E For sources subject to Regulation Number 3, Part B, Section III.J.2, the owner or operator must design and implement an approved source-specific monitoring plan (SSMP) to monitor emissions of nitrogen dioxide, PM_{2.5}, benzene, toluene, ethylbenzene, and xylene in accordance with Regulation Number 3, Part B, Section



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

III.J.2. The SSMP must be submitted for Division review and approval at least sixty (60) days prior to commencement of operation under this general permit. Failure to submit an approvable SSMP could result in revocation of the general permit registration. The owner or operator may discontinue source-specific monitoring requirements upon the Division's approval of a petition submitted in accordance with Regulation Number 3, Part B, Section III.J.2.f. (Regulation Number 3, Part B, Sections III.E. & III.G.7.c.)

V.F On or before June 30th of each calendar year the owner or operator of well production facilities in Colorado must submit a single annual report that includes actual emissions and specified information in the Division-approved report format. The information included in the annual report must be in accordance with the general reporting requirements of Regulation Number 7, Part B, Section V.B. Each calendar year owners or operators must maintain the information according to Regulation Number 7, Part D, Section V.C. for inclusion in the annual report. (Regulation Number 7, Part B, Section V)

V.G Unless otherwise defined in this general permit, any terms and conditions in this general permit are as defined per the most current version of Regulation Number 7, Part B, Section II.A.

V.H Storage Tanks

V.H.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections I.C., I.D., I.E., and I.F., including, but not limited to, compliance deadlines, control, monitoring, performance testing, inspection, recordkeeping, and reporting requirements.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- V.H.2 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections, II.B. and II.C., including, but not limited to, compliance deadlines, control, monitoring, performance testing, inspection, design, recordkeeping and reporting requirements.
- V.H.3 Storage tanks subject to 40 CFR Part 60, NSPS Subpart OOOO - Standards of Performance for Crude Oil and Natural Gas Facilities - must comply with all applicable requirements of that subpart including, but not limited to, emissions standards, testing, notification, monitoring, recordkeeping and reporting.
- V.H.4 Compliance with Conditions V.H.1 through V.H.3 fulfills the requirements to comply with Reasonably Available Control Technology (RACT) of Regulation Number 3, Part B, Section III.D.2.

V.I Hydrocarbon liquid loading operations

- V.I.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections I.C. and I.M., including, but not limited to, compliance deadlines, control, monitoring, inspection, design, recordkeeping and reporting requirements.
- V.I.2 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections II.B. And II.C.5., including, but not limited to, compliance deadlines, control, training, monitoring, inspection, design, recordkeeping and reporting requirements.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

V.I.3 Compliance with Conditions V.I.1 and V.I.2 fulfills the requirements to comply with RACT of Regulation Number 3, Part B, Section III.D.2.

V.J Gas Venting from Separators

V.J.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections II.B. and II.F., including, but not limited to, compliance deadlines, control, performance testing, and collection requirements.

V.J.2 Compliance with Condition V.J.1 fulfills the requirements to comply with RACT of Regulation Number 3, Part B, Section III.D.2.

V.K Routine or Predictable Emissions

V.K.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections II.G., including, but not limited to, compliance deadlines, control, monitoring, inspection, design, recordkeeping and reporting requirements.

V.K.2 The owner or operator must use best management practices to minimize hydrocarbon emissions during, or to reduce the frequency of routine or predictable gas venting emissions events. The owner and operator must develop and maintain a description of best management practices implemented to minimize hydrocarbon emissions during, or to reduce the frequency of, routine or predictable gas venting events covered by this general permit. The description must also include how those best management practices are implemented and performed. The description must be made available for Division review upon request. (Regulation Number 3, Part B, Section III.E.)



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- V.K.3 The owner or operator must perform routine or predictable gas venting emissions activities between October 1 and April 30, when practicable. (Regulation Number 3, Part B, Section III.E.)
- V.K.4 The owner or operator must avoid performing routine or predictable gas venting emissions activities on Ozone Action Days, when practicable. Ozone Action Day advisories are published by the Colorado Department of Public Health and Environment. The owner or operator shall maintain a record of all venting events occurring during Ozone Action Days. (Regulation Number 3, Part B, Section III.E.)
- V.K.5 The owner or operator, or authorized third party representative, must: (Regulation Number 3, Part B, Section III.E.)
- V.K.5.a Remain on site while a storage tank thief hatch is open if there are liquids in the tank, if the thief hatch was opened to perform routine or predictable emitting activities; and
 - V.K.5.b Minimize the duration of time the thief hatch is open during activities to determine the quality and/or quantity of storage tank contents.
- V.K.6 For equipment blowdowns, the owner or operator must avoid blowing down the equipment completely to ambient pressure, where practicable. (Regulation Number 3, Part B, Section III.E.)
- V.K.7 For pigging related activities, the owner or operator must install, operate and maintain a liquids handling system to remove any accumulated liquid prior to pig receiver depressurization. (Regulation Number 3, Part B, Section III.D.2.)



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

V.K.8 Compliance with Conditions V.K.1 through V.K.7 fulfills the requirements to comply with RACT of Regulation Number 3, Part B, Section III.D.2.

V.L Air Pollution Control Equipment used to control Storage Tanks, Hydrocarbon Liquids Loadout, Gas Venting from Separators, or Routine or Predictable Emissions

V.L.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections I.C., I.D., I.E., I.F. and I.M., including, but not limited to, compliance deadlines, control, training, monitoring, performance testing, inspection, design, recordkeeping and reporting requirements.

V.L.2 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Section II.B., II.C., II.F. and II.G., including, but not limited to, compliance deadlines, control, monitoring, performance testing, inspection, design, recordkeeping and reporting requirements.

V.M Natural gas-fired reciprocating internal combustion engines

V.M.1 Engines registered to operate in accordance with the provisions of this general permit must be subject to and must not exceed the emissions standards contained in *Table 1 - Emission Standards (g/hp-hr) for Rich Burn and Lean Burn Engines*. If the engine is subject to a more stringent applicable performance standard for a specific pollutant, the most stringent performance standard shall apply (Regulation Number 3, Part B, Section III.E).



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Table 1 - Emission Standards (g/hp-hr) for Rich Burn and Lean Burn Engines

Engine Design (horsepower (hp))	NOx	CO	VOC
Rich Burn (Greater than or equal to 100 and less than 500)	2.0	4.0	1.0
Lean Burn (Greater than or equal to 100 and less than 500)	2.0	4.0	1.0
Rich Burn (Greater than or equal to 500)	0.7	1.5	1.0
Lean Burn (Greater than or equal to 500)	2.0	1.5	0.7

V.M.2 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 26, Part B, Sections I.A. through I.D. including, but not limited to, compliance deadlines, control, emission standards, training, monitoring, performance testing, inspection, design, recordkeeping and reporting requirements.

V.M.3 Compliance with Conditions V.M.1 and V.M.2 fulfills the requirements to comply with RACT of Regulation Number 3, Part B, Section III.D.2.

V.N Diesel fuel-fired reciprocating internal combustion engines

V.N.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 26, Part B, Section I.D., including, but not limited to, compliance deadlines, control, emission



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

standards, training, monitoring, performance testing, inspection, design, recordkeeping and reporting requirements.

V.N.2 If an engine is subject to the New Source Performance Standards requirements of Regulation Number 6, Part A, Subpart IIII, Standards of Performance for Stationary Compression Ignition Internal Combustion Engines (CI ICE):

V.N.2.a Owners and operators of 2007 model year and later non-emergency stationary CI ICE with a displacement of less than 30 liters per cylinder must comply with the emission standards for new CI engines in 40 CFR 60.4201 for their 2007 model year and later stationary CI ICE, as applicable.

V.N.2.a.(i) All fuel used shall meet the following specifications:

V.N.2.a.i.(A) Sulfur content shall not exceed 15 ppm.

V.N.2.a.i.(B) Have a minimum cetane index of 40 or have a maximum aromatic compound content of 35% by volume.

Compliance shall be demonstrated by maintaining copies of the fuel specifications provided by the supplier on-site or in a readily accessible location and made available to the Division for inspection upon request.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- V.N.3 All engines and control devices must be installed, configured, operated, and maintained according to the specifications and instructions provided by the engine manufacturer.
- V.N.4 If the engine is equipped with a diesel particulate filter, the filter must be installed with a backpressure monitor that notifies the owner or operator when the high backpressure limit of the engine is approached. Records shall be kept of any corrective action taken after the backpressure monitor has notified the owner or operator that the high backpressure limit is approached. If the engine is an emergency stationary CI internal combustion engine that does not meet the standards applicable to non-emergency engines, the owner or operator must install a non-resettable hour meter prior to startup of the engine per Subpart IIII §60.4209(a).
- V.N.5 The following requirements of Regulation Number 6, Part A, Subpart A, General Provisions, including but not limited to, apply:
- V.N.5.a At all times, including periods of start-up, shutdown, and malfunction, the facility and control equipment shall, to the extent practicable, be maintained and operated in a manner consistent with good air pollution control practices for minimizing emissions. Determination of whether or not acceptable operating and maintenance procedures are being used will be based on information available to the Division, which may include, but is not limited to, monitoring results, opacity observations, review of operating and maintenance procedures, and inspection of the source. (Regulation Number 6, Part A. General Provisions from 40 CFR 60.11(d))



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

V.N.5.b No article, machine, equipment or process shall be used to conceal an emission which would otherwise constitute a violation of an applicable standard. Such concealment includes, but is not limited to, the use of gaseous diluents to achieve compliance with an opacity standard or with a standard which is based on the concentration of a pollutant in the gases discharged to the atmosphere. (Regulation Number 6, Part A. General Provisions from 40 CFR 60.12)

V.N.5.c Written notification of construction and initial startup dates shall be submitted to the Division as required under 40 CFR § 60.7.

V.N.5.d Records of startups, shutdowns, and malfunctions shall be maintained, as required under 40 CFR § 60.7.

V.N.6 Compliance with Conditions V.N.1 through V.N.5 fulfills the requirements to comply with RACT of Regulation Number 3, Part B, Section III.D.2.

V.O Fugitive Component leak emissions

V.O.1 The owner or operator must comply with all applicable requirements per the most current version of Regulation Number 7, Part B, Sections I.L., including, but not limited to, compliance deadlines, inspection, monitoring, repair, remonitoring, recordkeeping and reporting requirements.

V.O.2 The owner or operator must comply with all applicable per the most current version of Regulation Number 7, Part B, Section II.E., including, but not limited to, compliance deadlines, inspection, monitoring, repair, remonitoring, recordkeeping and reporting requirements.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

V.O.3 Compliance with Conditions V.O.1 and V.O.2 fulfill the requirements to comply with RACT of Regulation Number 3, Part B, Section III.D.2.

VI. Control, Operating, and Maintenance Requirements

VI.A Storage Tanks

VI.A.1 If air pollution control equipment is used to comply with the emission limits of this general permit, it must collect and control emissions from each storage tank by routing emissions to air pollution control equipment that achieves a VOC control efficiency of 95%. If a combustion device is used, it must have a design destruction efficiency of at least 98% for VOC. (Regulation Number 3, Part B, Section III.E.)

VI.A.2 If air pollution control equipment is used to comply with the emission limits of this general permit, the owner or operator must comply with the applicable requirements in Condition VI.E. (Regulation Number 3, Part B, Section III.E.)

VI.B Hydrocarbon Liquids Loadout

VI.B.1 If air pollution control equipment is used to comply with the emission limits of this general permit, it must achieve a VOC control efficiency of 95%. If a combustion device is used, it must have a design destruction efficiency of at least 98% for VOC. (Regulation Number 3, Part B, Section III.E.)

VI.B.2 If air pollution control equipment is used to comply with the emission limits of this general permit, the owner or operator must comply with the applicable requirements in Condition VI.E. (Regulation Number 3, Part B, Section III.E.)



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VI.C Gas Venting from Separators

VI.C.1 Gas coming off a separator, produced during normal operation from any oil and gas well, must either be routed to a gas gathering line or controlled by air pollution control equipment that achieves a VOC control efficiency of 95%. If a combustion device is used, it must have a design destruction efficiency of at least 98% for VOC. (Regulation Number 3, Part B, Section III.E.)

VI.C.2 If air pollution control equipment is used to comply with the emission limits of this general permit, the owner or operator must comply with the applicable requirements in Condition VI.E. (Regulation Number 3, Part B, Section III.E.)

VI.D Routine or Predictable Emissions

VI.D.1 If air pollution control equipment is used to comply with the emission limits of this general permit, it must achieve a VOC control efficiency of 95%. If a combustion device is used, it must have a design destruction efficiency of at least 98% for VOC. (Regulation Number 3, Part B, Section III.E.)

VI.D.2 If air pollution control equipment is used to comply with the emission limits of this general permit, the owner or operator must comply with the applicable requirements in Condition VI.E. (Regulation Number 3, Part B, Section III.E.)

VI.E Air Pollution Control Equipment used to control Storage Tanks, Hydrocarbon Liquids Loadout, Gas Venting from Separators, or Routine or Predictable Emissions

VI.E.1 All air pollution control equipment must be operated and maintained pursuant to the manufacturing specifications or equivalent to the extent



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

practicable, and consistent with technological limitations and good engineering and maintenance practices. The owner or operator must keep manufacturer specifications or equivalent on file. In addition, all such air pollution control equipment must be adequately designed and sized to achieve the control efficiency rates and to handle reasonably foreseeable fluctuations in emissions of VOCs and other hydrocarbons during normal operations. Fluctuations in emissions that occur when the separator dumps into the tank are reasonably foreseeable. (Regulation Number 3, Part B, Section III.E.)

- VI.E.2 If a combustion device is used to control emissions of VOCs and other hydrocarbons, it must be enclosed, have no visible emissions during normal operation, and be designed so that an observer can, by means of visual observation from the outside of the enclosed combustion device, or by other means approved by the Division, determine whether it is operating properly. (Regulation Number 3, Part B, Section III.E.)
- VI.E.3 All combustion devices used to control emissions of hydrocarbons must be equipped with and operate an auto-igniter. (Regulation Number 3, Part B, Section III.E.)
- VI.E.4 Air pollution control equipment must achieve a VOC control efficiency of 95% and must have a design destruction efficiency of at least 98% for VOC. (Regulation Number 3, Part B, Section III.E.)
- VI.E.5 The owner or operator must conduct visual inspections of air pollution control equipment. Visual inspections must follow the schedule in *Table 2 - ECD monitoring frequency*. Visual inspections must include, at a minimum: (Regulation Number 3, Part B, Section III.E.)



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- VI.E.5.a Inspection or monitoring of each combustion device to ensure that it is operating, including that the pilot light is lit and the auto-igniter is properly functioning. The owner or operator may use a surveillance system to maintain records on combustion device operation. For storage tanks equipped with a surveillance system or other Division-approved monitoring system, the owner or operator must check weekly that the system is functioning properly and that necessary information is being collected. Any loss of data or failure to collect required data must be treated as control device downtime and will be considered by the Division as if the data was not collected.
- VI.E.5.b Inspection or monitoring of each combustion device to ensure that the valves for the piping of gas to the pilot light are open and functioning properly.
- VI.E.5.c Inspection or monitoring of each combustion device to ensure the burner tray is not visibly clogged.
- VI.E.5.d Inspection of each combustion device for the presence or absence of smoke. If smoke is observed, either the equipment must be immediately shut-in to investigate the potential cause for smoke and perform repairs, as necessary, or EPA Method 22 must be conducted to determine whether visible emissions are present for a period of at least one (1) minute in fifteen (15) minutes.
- VI.E.5.e Inspection or monitoring of each vapor recovery unit to ensure that the unit is operating and that vapors are being routed to the unit.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- VI.E.5.f Inspection or monitoring of air pollution control equipment to ensure that valves for the piping of gas to the air pollution control equipment are open.
- VI.E.5.g Recording the flow meter readings or pressure actuator system data. For flow meter readings, this must include the maximum and minimum measured flow rate since the previous visual inspection. For pressure actuator system data, this must include the maximum and minimum measured pressures while the actuator valve is open since the previous visual inspection. An owner or operator may use automation to continuously record flow and/or pressure to the enclosed combustion devices(s).
- VI.E.5.h Visual inspections of air pollution control equipment must follow the schedule below:

Table 2 - ECD monitoring frequency

Area classification	Permitted facility-wide VOC emissions	Monitoring Frequency
Attainment	Less than 90 TPY	Weekly
Attainment	Greater than or equal to 90 TPY	Daily
Marginal/Moderate Non-	Less than 90 TPY	Weekly



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Area classification	Permitted facility-wide VOC emissions	Monitoring Frequency
Attainment		
Marginal/Moderate Non-Attainment	Greater than or equal to 90 TPY	Daily
Serious Non-Attainment	Less than 40 TPY	Weekly
Serious Non-Attainment	Greater than or equal to 40 TPY	Daily
Severe Non-Attainment	Less than 15 TPY	Weekly
Severe Non-Attainment	Greater than or equal to 15 TPY	Daily
Extreme Ozone Non-attainment	Any emissions	Daily

VI.E.5.i

VI.F Natural gas-fired reciprocating internal combustion engines

VI.F.1 This Condition VI.F applies only to engines covered by this general permit that are located at a stationary source that is synthetic minor or major for



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

NO_x, CO, or formaldehyde except as indicated in Condition VI.F.5.f.
(Regulation Number 3, Part B, Section III.E.)

VI.F.2 These general requirements apply to all parts of Condition VI.F: (Regulation Number 3, Part B, Section III.E.)

VI.F.2.a Each engine and associated catalytic control device, if applicable, must be maintained and operated according to the manufacturer's scheduled maintenance guide or equivalent. A company may establish its own equivalent maintenance schedule providing it achieves efficient engine operation and meets or exceeds the performance standards of this general permit.

VI.F.2.b A copy of maintenance schedules must be kept.

VI.F.2.c Records indicating the date and description of maintenance must be kept.

VI.F.2.d If maintenance activities or actions are dependent upon hours of operation, then engine operating hours must be tracked and recorded. If a method other than hours of operation is used, then the tracking method must be recorded.

VI.F.3 For each rich burn engine controlled with a Non-Selective Catalytic Reduction (NSCR) converter the following conditions must be met:
(Regulation Number 3, Part B, Section III.E.)

VI.F.3.a The engine must be equipped with an air-fuel ratio controller (AFRC).



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- VI.F.3.b Each AFRC must be maintained per the manufacturer's recommended maintenance or equivalent, including replacement of the oxygen sensor as necessary for oxygen sensor-based controllers. In the absence of specific manufacturer's recommendations or equivalent, the oxygen sensor must be replaced quarterly (or within a 2190-hour operating period if the engine does not run continuously).
- VI.F.4 For each lean burn engine equipped with a NO_x sensor the operator must follow the manufacturer's recommended maintenance schedule or equivalent. (Regulation Number 3, Part B, Section III.E.)
- VI.F.5 For each engine equipped with a catalytic emission control device, the following conditions must be met: (Regulation Number 3, Part B, Section III.E.)
- VI.F.5.a Engines subject to the requirements of Regulation Number 26, Part B, Section I.D.5. must comply with the monitoring requirements listed in Regulation Number 26, Part B, Section I.D.5.e.
- VI.F.5.b Engines subject to the requirements of Regulation Number 26, Part B, Section I.D.6. must comply with the monitoring requirements listed in Regulation Number 26, Part B, Section I.D.6.d.
- VI.F.5.c The catalyst must be cleaned, reconditioned and replaced per the manufacturer's recommended maintenance or equivalent.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VI.F.5.d The catalyst inlet temperature must be recorded weekly, unless a more stringent requirement is required in Regulation Number 26, Part B, Section I.D.5.e.(ii). If the temperature is outside of the range specified in this condition, corrective action must be taken and documented:

VI.F.5.d.(i) For a rich burn engine with an NSCR emission control device, the temperature into the catalyst must be between 750°F and 1250°F.

VI.F.5.d.(ii) For a lean burn engine with an oxidation catalyst emission control device, the temperature into the catalyst must be between 450°F and 1350°F.

VI.F.5.e The pressure drop across the catalyst bed must be recorded monthly. The monthly pressure drop values must not deviate by more than +/- 2 inches of water column of the most recent baseline value established according to this condition.

VI.F.5.e.(i) If the pressure drop value is out of range, the owner or operator must perform any one or more of the following:

VI.F.5.e.i.(A) Corrective action must be taken and documented, or

VI.F.5.e.i.(B) The owner or operator may artificially load the engine to within +/- 10 percent of the most recent baseline load value established. If



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

the pressure drop is then within range, the pressure drop and load values must be documented and no further corrective action is required, or

VI.F.5.e.i.(C) The owner or operator may forego maintenance activities and perform a portable analyzer test to establish a new pressure drop baseline.

VI.F.5.e.(ii) The baseline pressure drop must be established by recording the engine load and pressure drop across the catalyst during engine testing. The baseline pressure drop must be periodically reestablished as follows:

VI.F.5.e.ii.(A) During each initial compliance test required by the Initial and Periodic Compliance Test Requirements section of this general permit.

VI.F.5.e.ii.(B) During each periodic performance test required by this general permit.

VI.F.5.e.ii.(C) Within the first thirty (30) days of engine operation from the date that the catalyst is cleaned, reconditioned, or replaced.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VI.F.5.f For each engine equipped with a catalytic emission control device, periodic performance tests for NO_x and CO using a portable analyzer are required to indicate continued performance of the catalyst and proper operation of the engine following the schedule:

- VI.F.5.f.(i) Major sources of NO_x or CO allowed to operate under this general permit pursuant to Condition I.F must perform performance tests quarterly. The frequency must remain quarterly regardless of the number of consecutive passing tests.
- VI.F.5.f.(ii) Minor sources and synthetic minor sources with facility-wide permitted NO_x or CO emissions greater than or equal to the monitoring thresholds in *Table 3 - Portable analyzer testing NO_x and CO thresholds* must perform performance tests quarterly. After four (4) consecutive quarterly passing tests, the operator may perform performance tests semi-annually.
- VI.F.5.f.(iii) Minor sources and synthetic minor sources with facility-wide NO_x or CO permitted emissions lower than the monitoring thresholds in *Table 3 - Portable analyzer testing NO_x and CO thresholds* must perform performance tests semi-annually. After two (2) consecutive semi-annual passing tests the operator may perform performance tests annually.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Table 3 - Portable analyzer testing NOx and CO thresholds

Area classification	NOx monitoring threshold	CO monitoring threshold
Attainment	80	80
Marginal/Moderate Non-Attainment	80	Not Applicable
Serious Non-Attainment	40	Not Applicable
Severe Non-Attainment	15	Not Applicable
Extreme Ozone Non-attainment	Zero (0)	Not Applicable

VI.F.5.f.(iv) Each performance test should be performed with the engine operating at the highest load practicable. However, it is not required to artificially load the engine in order to perform the test.

VI.F.5.f.(v) All performance tests using a portable analyzer, including tests triggered by a replacement engine authorized by an Alternative Operating Scenario (AOS) in this general permit, must meet the



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

requirements established in the Portable Analyzer Monitoring Protocol or other Division approved protocol.

- VI.F.5.f.(vi) At the owner or operator's discretion, a performance test using a reference test method may be used instead of a portable analyzer.

VI.G Diesel fuel-fired reciprocating internal combustion engines

- VI.G.1 This Condition VI.G applies only to engines covered by this general permit that are located at a stationary source that is synthetic minor or major for NO_x, CO, or formaldehyde except as indicated in Condition VI.G.6. (Regulation Number 3, Part B, Section III.E.).
- VI.G.2 Each engine and associated emission control device, if applicable, must be maintained and operated according to the manufacturer's scheduled maintenance guide or equivalent. A company may establish its own equivalent maintenance schedule providing it achieves efficient engine operation and meets or exceeds the performance standards of this general permit. (Regulation Number 3, Part B, Section III.E.)
- VI.G.3 A copy of maintenance schedules must be kept. (Regulation Number 3, Part B, Section III.E.)
- VI.G.4 Records indicating the date and description of maintenance must be kept. (Regulation Number 3, Part B, Section III.E.)
- VI.G.5 If maintenance activities or actions are dependent upon hours of operation, then engine operating hours must be tracked and recorded. If a method



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

other than hours of operation is used, then the tracking method must be recorded. (Regulation Number 3, Part B, Section III.E.)

VI.G.6 For each engine, periodic performance tests for NO_x and CO using a portable analyzer are required to indicate continued proper performance and operation of the engine following the schedule (Regulation Number 3, Part B, Section III.E.):

VI.G.6.a Major sources of NO_x or CO allowed to operate under this general permit pursuant to Condition I.F must perform performance tests quarterly. The frequency must remain quarterly regardless of the number of consecutive passing tests.

VI.G.6.b Minor sources and synthetic minor sources with facility-wide permitted NO_x or CO emissions greater than or equal to the monitoring thresholds in *Table 4 - Portable analyzer testing NO_x and CO thresholds* must perform performance tests quarterly. After four (4) consecutive quarterly passing tests, the operator may perform performance tests semi-annually.

VI.G.6.c Minor sources and synthetic minor sources with facility-wide permitted NO_x or CO emissions lower than the monitoring thresholds in *Table 4 - Portable analyzer testing NO_x and CO thresholds* must perform performance tests semi-annually. After two (2) consecutive semi-annual passing tests the operator may perform performance tests annually.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Table 4 - Portable analyzer testing NOx and CO thresholds

Area classification	NOx monitoring threshold	CO monitoring threshold
Attainment	80	80
Marginal/Moderate Non-Attainment	80	Not Applicable
Serious Non-Attainment	40	Not Applicable
Severe Non-Attainment	15	Not Applicable
Extreme Ozone Non-attainment	Zero (0)	Not Applicable

VI.G.6.d Each performance test should be performed with the engine operating at the highest load practicable. However, it is not required to artificially load the engine in order to perform the test.

VI.G.6.e All performance tests using a portable analyzer, including tests triggered by a replacement engine authorized by an AOS in this general permit, must meet the requirements established in the Portable Analyzer Monitoring Protocol or other Division approved protocol.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VI.G.6.f At the owner or operator's discretion, a performance test using a reference test method may be used instead of a portable analyzer.

VII. Initial and Periodic Compliance Test Requirements

VII.A Storage Tanks

VII.A.1 Condensate, crude oil, or intermediate hydrocarbons: Within one hundred and eighty days (180) after commencement of operation under the terms of this general permit, and periodically following the schedule in *Table 5 - Sampling frequency*, the owner or operator must complete site-specific sampling including a compositional analysis of the pre-flash pressurized condensate, crude oil, or intermediate hydrocarbons routed to these storage tanks and, if necessary for emission factor development, a sales oil analysis to determine RVP and API gravity. Sampling and analysis must be in accordance with the guidance contained in PS Memo 25-02 and 17-01, and the pre-flash pressurized condensate, crude oil, or intermediate hydrocarbons sample must be obtained from the outlet of the low-pressure side of the separator immediately upstream of the storage tank. Results of the analysis must be used to calculate site-specific emissions factors for VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane (in units of lb/bbl of condensate, crude oil or intermediate hydrocarbons throughput) using Division approved methods. The total calculated emission factors must include flash, working, and breathing emissions. The owner or operator must use the results of this analysis to calculate actual emissions and to demonstrate compliance with emission limits starting with the calendar month following the sampling event. The owner or operator must report initial and periodic emission



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

factors as required per the most current version of Regulation Number 7, Part B, Section V. Records of initial and periodic site-specific sampling must be maintained by the owner or operator for a minimum of five (5) years and made available to the Division for inspection upon request. Owners or operators of condensate or crude oil storage tanks with uncontrolled actual VOC emissions below 80 tons per year and 20 tons per year, respectively, and that are also using state-default emission factors, are not required to comply with this condition.

- VII.A.2 Produced water: Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 5 - Sampling frequency*, the owner or operator must complete site-specific sampling including a compositional analysis of the pre-flash pressurized produced water routed to these storage tanks and, if necessary for emission factor development, a sales oil analysis to determine RVP and API gravity of the hydrocarbon liquids present in the produced water. Sampling and analysis must be in accordance with the guidance contained in PS Memo 25-02 and the pre-flash pressurized produced water sample must be obtained from the outlet of the low-pressure side of the separators immediately prior to the storage tank. Results of the analysis must be used to calculate site-specific emissions factors for VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane (in units of lb/bbl of produced water throughput) using Division approved methods. The total calculated emission factors must include flash, working and breathing emissions. The owner or operator must use the results of this analysis to calculate actual emissions and to demonstrate compliance with emission limits starting with the calendar month following the sampling event. The owner or operator must report



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

initial and periodic emission factors as required per the most current version of Regulation Number 7, Part B, Section V. Records of initial and periodic site-specific sampling must be maintained by the owner or operator for a minimum of five (5) years and made available to the Division for inspection upon request. Owners or operators using state-default emission factors are not required to comply with this condition.

VII.B Gas Venting from Separators

VII.B.1 Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 5 - Sampling frequency*, the owner or operator must complete a site specific extended gas analysis ("Analysis") including a compositional analysis of gas vented from this emissions unit. The gas sample must be obtained from the outlet of each distinct separator point covered by this general permit. Results of the Analysis must be used to calculate site-specific emissions factors for VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane (in units of lb/MMscf of gas vented throughput) using Division approved methods. The owner or operator must use the results of this Analysis to calculate actual emissions and to demonstrate compliance with emission limits starting with the calendar month following the sampling event. The owner or operator must report initial and periodic emission factors as required per the most current version of Regulation Number 7, Part B, Section V. Records of initial and periodic site-specific sampling must be maintained by the owner or operator for a minimum of five (5) years and made available to the Division for inspection upon request.

VII.C Routine or Predictable Emissions



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VII.C.1 When using a volumetric processes as allowed by the by the Annual Process Limitations section of this general permit: Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 5 - Sampling frequency*, the owner or operator must complete an site specific extended gas analysis ("Analysis"), of the gas vented during these activities in order to verify the VOC (weight fraction) of this emission stream. The gas sample must be obtained either from a gas line with VOC and HAPs contents that are representative of the gas vented during blowdown events or directly from the equipment blowdown line. Results of the Analysis must be used to calculate site-specific emission factors for VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane (in units of lb/MMSCF gas vented) using Division approved methods. For the pollutants identified in this permit condition as needing sampling, if any site-specific emission factors developed through this analysis are greater than the emissions factors listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the Air Pollutant Emission Notices (APEN) included with the general permit registration, the owner or operator must use the results of this analysis to calculate actual emissions and to demonstrate compliance with emission limits starting with the calendar month following the sampling event. The owner or operator must report initial and periodic emission factors as required per the most current version of Regulation Number 7, Part B, Section V. Records of initial and periodic site-specific sampling must be maintained by the owner or operator for a minimum of five (5) years and made available to the Division for inspection upon request.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VII.C.2 When using events-based processes as allowed by the by the Annual Process Limitations section of this general permit: Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 5 - Sampling frequency*, the owner or operator must complete an analysis of all equipment associated with these activities. For each activity, the owner or operator must indicate the following for each piece of equipment:

VII.C.2.a Actual unique physical volume between isolation valves, as built.

The owner or operators may use engineering calculations and estimations to determine the actual unique physical volume between isolation valves (i.e. internal physical volume);

VII.C.2.b Absolute pressure at actual conditions in the unique physical volume. The owner or operators may use engineering calculations and estimations to determine the absolute pressure.

VII.C.2.c The owner or operator must complete either:

VII.C.2.c.(i) a site-specific extended gas analysis of the gas vented during this activity in order to verify the VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane content (weight fraction) of this emission stream. The gas sample must be obtained either from a gas line with VOC and HAPs contents that are representative of the gas vented during blowdown events or directly from the equipment blowdown line.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- VII.C.2.c.(ii) a site-specific sampling including a compositional analysis of the pre-flash pressurized condensate, crude oil, intermediate hydrocarbons, or produced water routed to storage tanks and, if necessary for emission factor development, a sales oil analysis to determine RVP and API gravity in order to verify the VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane content (weight fraction) of this emission stream.

Results of the analysis must be used to calculate site-specific emission factors for VOC, benzene, toluene, ethylbenzene, xylenes, n-hexane, and 2,2,4-trimethylpentane using Division approved methods. For the pollutants identified in this permit condition as needing sampling, if any site-specific emission factors developed through this analysis are greater than the emissions factors listed in the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the Air Pollutant Emission Notices (APEN) included with the general permit registration, the owner or operator must use the results of this analysis to calculate actual emissions and to demonstrate compliance with emission limits starting with the calendar month following the sampling event. The owner or operator must report initial and periodic emission factors as required per the most current version of Regulation Number 7, Part B, Section V. Records of initial and periodic site-specific sampling must be maintained by the owner or operator for a minimum of five (5) years and made available to the Division for inspection upon



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

request. Owners or operators using state-default emission factors are not required to comply with this condition.

Table 5 - Sampling frequency

Area classification	Facility-wide permitted VOC emissions	Sampling Frequency	Sampling frequency for storage tanks with actual uncontrolled VOC emissions $\geq$ 80 tons per year.
Attainment	Less than 90 TPY	Every five years	Every two years
Attainment	Greater than or equal to 90 TPY	Annually	Annually
Marginal/Moderate Non-Attainment	Less than 90 TPY	Every five years	Every two years
Marginal/Moderate Non-Attainment	Greater than or equal to 90 TPY	Annually	Annually
Serious Non-Attainment	Less than 40 TPY	Every five	Every two years



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Area classification	Facility-wide permitted VOC emissions	Sampling Frequency	Sampling frequency for storage tanks with actual uncontrolled VOC emissions $\geq$ 80 tons per year.
		years	
Serious Non-Attainment	Greater than or equal to 40 TPY	Annually	Annually
Severe Non-Attainment	Less than 15 TPY	Every five years	Every two years
Severe Non-Attainment	Greater than or equal to 15 TPY	Annually	Annually
Extreme Ozone Non-attainment	Any emissions	Annually	Annually

VII.D Enclosed Combustion Devices

VII.D.1 Within one hundred and eighty days (180) after commencement of operation under the terms of this general permit, the owner or operator must demonstrate compliance with opacity standards, using EPA Reference



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Method 22, 40 C.F.R. Part 60, Appendix A, on basis to determine the presence or absence of visible emissions. “Visible Emissions” means observations of smoke for any period or periods of duration greater than or equal to one minute in any fifteen-minute period during normal operation. (Regulation Number 7, Part B, Sections II.B.2. and II.A.45.)

VII.D.2 For sources subject to Conditions VI.A and VI.C, the owner or operator must conduct source performance tests to demonstrate compliance with the requested destruction efficiency of 95% for volatile organic compounds (VOC). The owner or operator must comply with the following testing schedules:

VII.D.2.a If the enclosed combustion device is subject to Regulation Number 7, Part B, Section II.B, the owner or operator must follow the source performance testing schedule established in the most current version of Regulation Number 7, Part B, Section II.B.2.h.

VII.D.2.b If the enclosed combustion device is NOT subject to Regulation Number 7, Part B, Section II.B, the owner or operator must complete an initial source performance test within one hundred and eighty days (180) after commencement of operation under the terms of this general permit, and periodically every five (5) years.

Performance tests are not required for enclosed combustion devices serving solely as limited-use control devices during vapor recovery unit downtime. A performance test that does not demonstrate that an enclosed combustion device is achieving at least 95% destruction efficiency for VOC is considered a failing test.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

The compliance test must be conducted in accordance with the APCD Compliance Test Manual (as updated and amended by the Division), including deadlines for preparation and submittal of the protocol for Division review and approval and for submittal of the test report. All compliance testing must be approved by the Division prior to conducting the test. Additional time may be granted upon written request. (Regulation Number 3, Part B, Section III.G.3.)

If the results of compliance tests do not demonstrate compliance with requested destruction efficiency of 95% the owner or operator must, within thirty (30) days, follow the manufacturer's repair instructions, if available, or best combustion engineering practices to return the device to compliant operation or shut-in all equipment or operations controlled by the enclosed combustion device.

Owners or operators must retest the enclosed combustion device within ninety (90) days of corrective action in response to a failed test or within thirty (30) days of return to operation if the equipment or operations controlled by the enclosed combustion device were shut-in as a response to a failed test. Division approval of the testing protocol is not required for a retest where: the owner or operator is following the same test protocol as the original, failed test, and conditions have not materially changed such that a new test protocol would be required.

VII.E Natural gas-fired reciprocating internal combustion engines

- VII.E.1 Engines subject to Condition IV.A.6.a: Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 6 - Sampling and testing*



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

frequency, the owner or operator must sample and analyze the heat content of the fuel used in the engines covered by this general permit using appropriate ASTM Methods or equivalent, if approved in advance by the Division. The fuel heat content must be based on the higher (gross) heating value (HHV) of the fuel. Results of the heat content verification must be retained for a period of five years and made available to the Division upon request. The heat content indicated by the most recent analysis must be used to monitor compliance with the emission limits for each engine.

VII.E.2 Within one hundred and eighty days (180) after commencement of operation under this general permit, the owner or operator must perform an initial compliance test on these emissions points to measure the emission rate(s) for the pollutants listed below in order to demonstrate compliance with the emission limits in the APEN(s) or approval letter. The compliance test must be conducted in accordance with the APCD Compliance Test Manual (as updated and amended by the Division), including deadlines for preparation and submittal of the protocol for Division review and approval and for submittal of the test report. All compliance testing must be approved by the Division prior to conducting the test. Any compliance test conducted to show compliance with a monthly or annual emission limitation shall have the results projected up to the monthly or annual averaging time by multiplying the test results by the allowable number of operating hours for that averaging time (Regulation Number 3, Part B, Section III.G.3.)

VII.E.2.a Oxides of Nitrogen using EPA approved methods.

VII.E.2.b Carbon Monoxide using EPA approved methods.

VII.E.2.c Volatile Organic Compounds using EPA approved methods.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VII.F Diesel fuel-fired reciprocating internal combustion engines

VII.F.1 Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 6 - Sampling and testing frequency*, the owner or operator must demonstrate compliance with opacity standards, using EPA Reference Method 9, 40 C.F.R. Part 60, Appendix A, to measure opacity from this source. The opacity must be measured and interpreted as an average of the readings taken over fifteen (15) second intervals for a total of six (6) minutes. (Regulation Number 1, Section II.A.1.)

VII.F.2 Within one hundred and eighty days (180) after commencement of operation under this general permit, the owner or operator must perform an initial compliance test on the engine to measure the emission rate(s) for the pollutants listed below in order to demonstrate compliance with the emission limits in the APEN or approval letter. The compliance test must be conducted in accordance with the APCD Compliance Test Manual (as updated and amended by the Division), including deadlines for preparation and submittal of the protocol for Division review and approval and for submittal of the test report. All compliance testing must be approved by the Division prior to conducting the test. Any compliance test conducted to show compliance with a monthly or annual emission limitation shall have the results projected up to the monthly or annual averaging time by multiplying the test results by the allowable number of operating hours for that averaging time (Regulation Number 3, Part B, Section III.G.3.)

VII.F.2.a Oxides of Nitrogen using EPA approved methods.

VII.F.2.b Carbon Monoxide using EPA approved methods.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VII.F.2.c Volatile Organic Compounds using EPA approved methods.

VII.G Fugitive component leak emissions

- VII.G.1 Within one hundred and eighty days (180) of commencement of operation under this general permit, the owner or operator must complete a physical hard count of facility components. This hard count must distinguish between components in “gas service”, “light liquid service”, “water/oil service”, and “heavy liquid service”. The owner or operator must use the most recent component count to determine actual emission and to show compliance with annual emission limits established in this general permit.
- VII.G.2 Any time a physical change is made at the facility which increases the number of components, and not to exceed a period of five years from the previous count, the operator must complete a physical hard count of facility components. This hard count must distinguish between components in “gas service”, “light liquid service”, “water/oil service”, and “heavy liquid service”. The hard-count and running total of all additions and subtractions to the component count must be maintained and made available for the Division upon request. The owner or operator must use the most recent component count to determine actual emission and to show compliance with annual emission limits established in this general permit.
- VII.G.3 Within one hundred and eighty days (180) after commencement of operation under this general permit, and periodically following the schedule in *Table 6 - Sampling and testing frequency*, the owner or operator must complete an extended gas analysis of gas samples and an extended natural gas liquids analysis of liquids that are representative of volatile organic compounds (VOC) and hazardous air pollutants (HAP) that may be released



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

as fugitive emissions. These extended gas and liquids analyses must be used in the compliance demonstration with emission limits. The owner or operator that assumes a VOC content of 100% for their gas service is not required to obtain extended gas analyses of gas samples. The owner or operator that assumes a VOC content of 100% for the heavy oil service, light oil service, and water/oil service is not required to obtain extended natural gas liquids analyses of liquids.

Table 6 - Sampling and testing frequency

Source type	Minor sources	Synthetic minor sources	Major sources allowed under Condition I.F
Natural gas-fired reciprocating internal combustion Engines	Every five years	Annually	Semi-annually
Diesel fuel-fired reciprocating internal combustion engines	Every five years	Annually	Semi-annually
Fugitive component leak emissions	Every five years	Annually	Annually



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VIII. Recordkeeping and Reporting Requirements

VIII.A All records required by the Recordkeeping and Reporting Requirements Section of this general permit must be kept for five (5) years. The owner or operator must make these records available to the Division upon request. (Regulation Number 3, Part B, Section III.E.)

VIII.B The following records must be maintained: (Regulation Number 3, Part B, Section III.E.)

VIII.B.1 The current issuance of this general construction permit.

VIII.B.2 The general permit registration approval letter, once received.

VIII.B.3 Records of commencement of operation for each individual piece of equipment covered under this general permit.

VIII.B.4 Records of commencement of operation for air pollution control equipment used to meet the requirements of this general permit.

VIII.C The owner or operator must maintain records of emission calculations performed to comply with the Annual Emission Limitations section in this general permit. (Regulation Number 3, Part B, Section III.E.)

VIII.D The owner or operator must maintain process records used to show compliance with the requirements in the Annual Process Limitations section of this general permit. (Regulation Number 3, Part B, Section III.E.)

VIII.E The owner or operator must maintain records of monitoring, inspections, tests, and observations performed to comply with the Control, Operating and Maintenance



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Requirements section in this general permit. (Regulation Number 3, Part B, Section III.E.)

VIII.F The owner or operator must maintain records of any initial or periodic testing performed to comply with the requirements in the Initial and Periodic Compliance Requirements section in this general permit. (Regulation Number 3, Part B, Section III.E.)

VIII.G The owner or operator must maintain records of the analysis used to demonstrate compliance with the National Ambient Air Quality Standards (NAAQS) submitted with the GP12 registration.

VIII.H The owner or operator of facilities not subject to Condition I.F must track and keep records of potential emissions from all insignificant activities at the stationary source on an annual basis to demonstrate compliance with facility emission limitations as indicated in *Table 7 - EPA Ozone Classification Thresholds* below. An inventory of each insignificant activity and associated emission calculations must be made available to the division for inspection upon request. For the purposes of this condition, insignificant activities are defined as any activity or equipment, which emits any amount but does not require an APEN or is permit exempt. Note that for reclassifications of an existing nonattainment area, the limit applies upon the effective date of reclassification. Total potential emissions from the facility, including all permitted emissions and potential to emit from all insignificant activities, must be less than: (Regulation Number 3, Part B, Section III.E.)

Table 7 - EPA Ozone Classification Thresholds



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Criteria Pollutant	Ozone attainment area (tons per year)	Moderate ozone non-attainment area (tons per year)	Serious ozone non-attainment area (tons per year)	Severe ozone non-attainment area (tons per year)	Extreme ozone non-attainment area (tons per year)
Volatile Organic Compounds (VOC)	100	100	50	25	10
Nitrogen Oxides (NOx)	100	100	50	25	10
Carbon Monoxide (CO)	100	100	100	100	100
Particulate Matter (PM)	100	100	100	100	100
Particulate Matter (PM10)	100	100	100	100	100



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Criteria Pollutant	Ozone attainment area (tons per year)	Moderate ozone non-attainment area (tons per year)	Serious ozone non-attainment area (tons per year)	Severe ozone non-attainment area (tons per year)	Extreme ozone non-attainment area (tons per year)
Sulfur Dioxide (SO ₂)	100	100	100	100	100
Individual HAPs	10	10	10	10	10
Total HAPs	25	25	25	25	25

VIII.I Storage Tanks, Hydrocarbon Loadout, Gas Venting from Separators, and Routine or Predictable Emissions

VIII.I.1 If air pollution control equipment is used to comply with the emission limits of this general permit, the owner or operator must comply with the requirements applicable requirements in Condition VIII.J. (Regulation Number 3, Part B, Section III.E.)

VIII.J Air Pollution Control Equipment used to control Storage Tanks, Hydrocarbon Liquids Loadout, Gas Venting from Separators, or Routine or Predictable Emissions

VIII.J.1 Except as specified in Condition VIII.J.1.g, the owner or operator must maintain records for a period of five (5) years and make them available to



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

the Division upon request, including: (Regulation Number 3, Part B, Section III.E.)

- VIII.J.1.a Records of the make, model, serial number or other unique identifier, and AIRS ID (if assigned) of each enclosed combustion device; associated facility name and location; and the range of gas flow or pressure at which the combustion device is designed to operate.
- VIII.J.1.b Records of visual inspections required by this general permit including the time and date of each inspection and a description of any problems observed, description and date of any corrective action(s) taken, and name of employee or third party performing corrective action(s).
- VIII.J.1.c Records of the date and result of any EPA Method 22 test or investigation.
- VIII.J.1.d Records of the date and duration of any period where the air pollution control equipment is not operating.
- VIII.J.1.e Monthly records of the total hours the vapor recovery unit is not operating, the total throughput volume, and total throughput volume during the time the vapor recovery unit is not operating.
- VIII.J.1.f Records of inlet gas flow rate.
- VIII.J.1.g Records of performance tests must be maintained for the life of the equipment that the enclosed combustion device is used to control (even if ownership or control of the device is transferred), including manufacturer model and serial number(s) of devices



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

tested; the date of the test; a copy of the test protocol followed; a certification by a responsible official that the performance test was conducted in accordance with a Division-approved test protocol; the enclosed combustion device parameters required by the test protocol; documentation of the methods and results of the test, including whether the device passed or failed and the tested control efficiency; and the date and description of any actions taken in response to a failed test.

VIII.J.1.h Records of flow meter or pressure actuation system calibration and maintenance conducted as required by this general permit, including manufacturer specifications and schedule if available.

VIII.K Natural gas-fired reciprocating internal combustion engines

VIII.K.1 The following records must be kept for a period of five years and made available to the Division upon request. (Regulation Number 3, Part B, Section III.E.)

VIII.K.1.a Records of performance tests conducted as required by this general permit, including the date, engine settings on the date of the test, and documentation of the methods and results of the testing.

VIII.K.1.b Records of periodic portable analyzer monitoring, including the date, engine settings on the date of the monitoring, and documentation of the results of the monitoring. These records must include any demonstration that no portable analyzer monitoring was required as allowed by this general permit, if applicable.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

VIII.K.1.c Records of catalyst monitoring required by this general permit and any actions taken to address monitored values outside the temperature or pressure drop parameters, including the date and a description of actions taken.

VIII.L Diesel fuel-fired reciprocating internal combustion engines

VIII.L.1 The following records must be kept for a period of five years and made available to the Division upon request. (Regulation Number 3, Part B, Section III.E.)

VIII.L.1.a Records of performance tests conducted as required by this general permit, including the date, engine settings on the date of the test, and documentation of the methods and results of the testing.

VIII.L.1.b Records of periodic portable analyzer monitoring, including the date, engine settings on the date of the monitoring, and documentation of the results of the monitoring. These records must include any demonstration that no portable analyzer monitoring was required as allowed by this general permit, if applicable.

IX. Alternative Operating Scenarios (AOS)

IX.A Storage Tanks

IX.A.1 Provided that there are no emissions increases and that the emission limits set forth in this general permit are still met, the owner or operator may invoke an AOS for the following changes to an existing tank battery



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

registered under the general permit without modifying the GP12 registration:

- IX.A.1.a Replacement of an existing control device by control equipment of the same make, model and no changes to the configuration that will impact a dispersion analysis; or
- IX.A.1.b Replacement of a storage vessel with another storage vessel of the same type, dimensions, and physical characteristics (e.g., same design, same color, same construction material) and no changes to the configuration that will impact a dispersion analysis; or
- IX.A.1.c Replacement of a separator with another separator of the same type, dimensions, and physical characteristics (e.g., same design) with no changes to the configuration that will impact a dispersion analysis.

- IX.A.2 The owner or operator must maintain a log to contemporaneously record the date and description of any change made under the provisions of this AOS. (Regulation Number 3, Part A, Section IV.A.1.)

IX.B Hydrocarbon Liquids Loadout

- IX.B.1 Provided that there are no emissions increases and that the emission limits set forth in this general permit are still met, the owner or operator may invoke an AOS for the following changes to an existing tank battery registered under the general permit without modifying the GP12 registration:



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

IX.B.1.a Replacement of an existing control device by control equipment of the same make, model, and no changes to the configuration that might impact a dispersion analysis.

IX.B.2 The owner or operator must record in a log the date and description of any change made under the provisions of this AOS. This log must be maintained contemporaneously with any changes made under the AOS (Regulation Number 3, Part A, Section IV.A.1.)

IX.C Gas Venting from Separators

IX.C.1 Provided that there are no emissions increases and that the emission limits set forth in this general permit are still met, the owner or operator may invoke an AOS for the following changes to an existing separator registered under the general permit without modifying the GP12 registration:

IX.C.1.a Replacement of an existing control device by control equipment of the same make, model and no changes to the configuration that will impact a dispersion analysis; or

IX.C.1.b Replacement of a separator with another separator of the same type, dimensions, and physical characteristics (e.g., same design) with no changes to the configuration that will impact a dispersion analysis.

IX.C.2 The owner or operator must record in a log the date and description of any change made under the provisions of this AOS. This log must be maintained contemporaneously with any changes made under the AOS (Regulation Number 3, Part A, Section IV.A.1.)

IX.D Enclosed Combustion Devices



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- IX.D.1 Provided that there are no emissions increases and that the emission limits set forth in this general permit are still met, the owner or operator may invoke an AOS for the following changes to an existing enclosed combustion device or a bank of enclosed combustion devices under the general permit without modifying the GP12 registration:
- IX.D.1.a Replacement of an existing control device by control equipment of the same make, model, and no changes to the configuration that might impact a dispersion analysis.
- IX.D.2 An AOS otherwise allowed by this general permit may not be used if the modification will result in a facility classification change from true minor to synthetic minor. In this case, a revised registration request must be submitted for review.
- IX.D.3 An AOS otherwise allowed by this general permit cannot be used for the permanent replacement of an entire combustion device at any source that has become a major stationary source for purposes of Prevention of Significant Deterioration or Non-Attainment Area New Source Review ("PSD/NANSR") unless the existing combustion device has emission limits that are below the significance levels in Regulation Number 3, Part D, II.A.44. Nothing in this AOS shall preclude the Division from taking an action, based on any permanent combustion device replacement(s), for circumvention of any state or federal PSD/NANSR requirement. Additionally, if any permanent combustion device replacement(s) constitute(s) a circumvention of applicable PSD/NANSR requirements, nothing in this AOS shall excuse the permittee from complying with PSD/NANSR and applicable permitting requirements.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- IX.D.4 An Air Pollutant Emissions Notice (APEN) that includes the specific manufacturer, model, and serial number of the replacement control device must be filed with the Division within 14 calendar days of commencing operation of a replacement control device under the AOS provision. The APEN must be accompanied by the appropriate APEN filing fee and a cover letter explaining that the owner or operator is exercising an AOS and has replaced the control device.
- IX.D.5 For sources subject to Conditions VI.A and VI.C, the owner or operator must conduct a source performance test to demonstrate compliance with the requested destruction efficiency of 95% for volatile organic compounds (VOC).
- IX.D.5.a If the enclosed combustion device is subject to Regulation Number 7, Part B, Section II.B, the owner or operator must follow the source performance testing schedule established in the most current version of Regulation Number 7, Part B, Section II.B.2.h.
- IX.D.5.b If the enclosed combustion device is NOT subject to Regulation Number 7, Part B, Section II.B, the owner or operator must complete an initial source performance test within one hundred and eighty days (180) of startup of the replacement control device in accordance with the AOS provisions.

Performance tests are not required for enclosed combustion devices serving solely as limited-use control devices during vapor recovery unit downtime. A performance test that does not demonstrate that an enclosed combustion device is achieving at least 95% destruction efficiency for VOC is considered a failing test. During the test, the owner or operator must measure. The test protocol, test, and test report must be in accordance with the



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

requirements of the Air Pollution Control Division Compliance Test Manual. The test protocol must include testing under all representative operating scenarios at the control device. All compliance testing must be approved by the Division prior to conducting the test. Additional time may be granted upon written request. (Regulation Number 3, Part B, Section III.G.3.)

If the results of compliance tests do not demonstrate compliance with requested destruction efficiency of 95% the owner or operator must, within thirty (30) days, follow the manufacturer's repair instructions, if available, or best combustion engineering practices to return the device to compliant operation or shut-in all equipment or operations controlled by the enclosed combustion device.

Owners or operators must retest the enclosed combustion device within ninety (90) days of corrective action in response to a failed test or within thirty (30) days of return to operation if the equipment or operations controlled by the enclosed combustion device were shut-in as a response to a failed test. Division approval of the testing protocol is not required for a retest where: the owner or operator is following the same test protocol as the original, failed test, and conditions have not materially changed such that a new test protocol would be required.

IX.E Natural gas-fired reciprocating internal combustion engines

IX.E.1 Provided that the emission limits set forth in Condition III are still met, the owner or operator may invoke an AOS for the following modifications to an existing engine registered under the general permit:

IX.E.1.a An existing registered engine may be temporarily replaced with a different or like-kind engine in accordance with Attachment A.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

IX.E.1.b An existing registered engine may be permanently replaced with a like-kind engine in accordance with Attachment A.

IX.E.1.b.(i) A permanent replacement engine must not include changes to the configuration that might impact a dispersion analysis, this includes but is not limited to:

IX.E.1.b.i.(A) Stack heights;

IX.E.1.b.i.(B) Stack locations; or

IX.E.1.b.i.(C) Exit velocity.

IX.E.1.c Routine maintenance, repair, or like-kind replacement of control equipment.

IX.E.1.d Addition of a control device where the operator will not claim federally enforceable credit for the emission reductions achieved.

IX.E.2 Modified or replacement equipment are subject to all requirements set forth in this general permit including, but not limited to, recordkeeping and reporting, and operation and maintenance plans.

IX.E.3 An AOS otherwise allowed by this general permit may not be used if the modification will result in a facility classification change from true minor to synthetic minor. In this case, a revised GP12 registration request must be submitted for review.

IX.F Diesel fuel-fired reciprocating internal combustion engines



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- IX.F.1 Provided that the emission limits set forth in Condition III are still met, the owner or operator may invoke an AOS for the following modifications to an existing engine registered under the general permit:
- IX.F.1.a An existing registered engine may be temporarily replaced with a different or like-kind engine in accordance with Attachment B.
- IX.F.1.b An existing registered engine may be permanently replaced with a like-kind engine in accordance with Attachment B.
- IX.F.1.b.(i) A permanent replacement engine must not include changes to the configuration that might impact a dispersion analysis, this includes but is not limited to:
- IX.F.1.b.i.(A) Stack heights;
- IX.F.1.b.i.(B) Stack locations; or
- IX.F.1.b.i.(C) Exit velocity.
- IX.F.1.c Routine maintenance, repair, or like-kind replacement of control equipment.
- IX.F.1.d Addition of a control device where the operator will not claim federally enforceable credit for the emission reductions achieved.
- IX.F.2 Modified or replacement equipment are subject to all requirements set forth in this general permit including, but not limited to, recordkeeping and reporting, and operation and maintenance plans.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

IX.F.3 An AOS otherwise allowed by this general permit may not be used if the modification will result in a facility classification change from true minor to synthetic minor. In this case, a revised GP12 registration request must be submitted for review.

X. General Permit Registration Approval Process

X.A Conditional certification of a registration under this general permit is effective from the date the registration is deemed complete as indicated in Condition X.B. A complete registration request consists of all general permit application materials required by the Division and associated fees. The owner or operator may commence construction and operation of the well production facility as represented in the registration once the registration is deemed complete. In the event the well production facility does not qualify for registration under the general permit the owner or operator accepts the liability of commencing these activities.

X.B The Division will determine completeness of the general permit registration request within sixty (60) days of the date of receipt of the request and provide written notification to the applicant. If the applicant does not receive notification of a completeness determination from the Division within sixty (60) days, the registration request will be deemed complete for the purpose of Conditions X.A and X.B.

X.C The Division has ninety (90) calendar days from the date the Division receives a complete general permit registration request to provide the applicant with a written approval or denial of the registration.

XI. General Permits Terms and Administration

XI.A A facility that becomes a major source for Title V Operating Permit requirements due to reclassification, and is operating under Condition I.F, must submit a Title V



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

operating permit application requesting to incorporate the terms of this general permit into a Title V Operating Permit. The due date for this application is one year from the date on which the facility becomes subject to Title V Operating Permit requirements.

XI.B In accordance with Colorado Regulation Number 3, Part C, Section V.C.5., some or all of the monitoring specified in this general permit will be incorporated as specific conditions in the source's Title V Operating Permit (if applicable). Additional monitoring not listed in this general permit may be included in the source's Operating Permit in order to satisfy the periodic monitoring requirements of Regulation Number 3, Part C, Section V.C.5.b.

XI.C General Terms

XI.C.1 A revised APEN must be filed: (Regulation Number 3, Part A, Section II.C.)

XI.C.1.a Annually by April 30th of the year following a significant increase in emissions as follows:

XI.C.1.a.(i) For VOC, a change in actual emissions of five (5) tons per year or more, above the level reported on the last APEN; or

XI.C.1.a.(ii) For VOC and NO_x sources in ozone non-attainment areas, a change in annual actual emissions of one ton per year or more or five percent, whichever is greater, above the level reported on the last APEN submitted to the Division; or

XI.C.1.a.(iii) For any non-criteria reportable pollutant, if the emissions increase by 50% or five (5) tons per



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

year, whichever is less, above the level reported on the last APEN.

XI.C.1.b Whenever there is a change in the owner or operator of any facility, process, or activity; or

XI.C.1.b.(i) As required by the requirements in the AOS section of this general permit,

XI.C.1.b.(ii) No later than 30 days before the existing APEN expires.

XI.C.2 The requirements of Regulation Number 3, Part D apply at such time that any well production facility or modification becomes a major well production facility or major modification solely by virtue of a relaxation in any enforceable limitation that was established after August 7, 1980 on the capacity of the well production facility or modification to otherwise emit a pollutant, such as a restriction on hours of operation. (Regulation Number 3, Part D, Sections V.A.7.b. and VI.B.4.)

XI.C.3 This well production facility may be able to utilize the Affirmative Defense Provision for Excess Emissions During Malfunctions contained in Common Provisions Regulation, Part II, Subpart E. The owner or operator must notify the Division of any malfunction condition which causes a violation of any emission limit or limits stated in this general permit as soon as possible, but no later than noon of the next working day, followed by written notice to the Division addressing all of the criteria set forth in Part II.E.1 of the Common Provisions Regulation.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- XI.C.4 This general permit is granted subject to all rules and regulations of the Commission, the Colorado Air Pollution Prevention and Control Act C.R.S. (25-7-101 et seq), and to those general and specific terms and conditions included in this document.
- XI.C.5 Unless specifically stated otherwise, the general and specific conditions contained in this general permit have been determined by the Division to be necessary to assure compliance with the provisions of section 25-7-114.5(7)(a), C.R.S.
- XI.C.6 Each and every condition of this general permit is a material part hereof and is not severable. Any challenge to or appeal of a condition herein by the permittee will constitute a rejection of the entire permit and upon such occurrence, this general permit must be deemed denied ab initio.
- XI.C.7 Section 25-7-114.7(2)(a), C.R.S. requires that all well production facilities required to file an APEN must pay an annual fee to cover the costs of inspections and administration.
- XI.C.8 Violation of the terms of a permit or of the provisions of the Colorado Air Pollution Prevention and Control Act or the regulations of the Commission may result in administrative, civil or criminal enforcement actions under Sections 25-7-115 (enforcement), -121 (injunctions), -122 (civil penalties), -122.1 (criminal penalties), C.R.S.
- XI.C.9 Registration under this general permit is approved in reliance upon the accuracy and completeness of information supplied by the owner or operator and is conditioned upon operation of the well production facility in accordance with this information and with representations made by the owner or operator or owner or operator's agents. It is valid only for the



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

equipment and operations or activity specifically identified on the general permit registration.

XI.D Registration Revision/Termination

- XI.D.1 The Division may deny or revoke registration under the general permit under the circumstances specified in Regulation Number 3, Part B, Section III.I.3.c. and require the owner or operator to apply for a source-specific construction permit as required by Regulation Number 3, Part B.
- XI.D.2 A registration under this general permit may be reissued to a new owner by the Division as provided in Regulation Number 3, Part B, Section II.B. upon request for transfer of ownership and the submittal of a revised APEN, revised general permit registration and the required fees.
- XI.D.3 Registration under this general permit is voluntary. The owner or operator may withdraw or cancel a registration under this general permit at any time by notifying the Division in writing. The owner or operator can apply for a source-specific construction permit as necessary per Regulation Number 3, Part B.

XI.E General Permit Revision/Termination

- XI.E.1 This general permit remains in effect until revised or terminated by the Division in accordance with the provisions of Regulation Number 3.
- XI.E.2 After public notice and comment as provided by Regulation Number 3, Part B, Section III.I.7., the Division may revise this general permit in order to add or delete requirements or limitations to the permit. This public notice must be conducted in a manner consistent with the provisions of Regulation Number 3, Part B, Section III.C.4.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- XI.E.3 If the Division revises this general permit, it will first provide written notice to affected owners or operators. The notice will advise the owner or operator that any existing registration to use the general permit will be automatically converted to a registration to use the revised general permit, provided that the owner or operator continues to meet all requirements of the revised general permit. Persons not wishing to continue coverage under the revised general permit have the option of applying for a source-specific construction permit as required by Regulation Number 3, Part B, or apply for a different general permit, as appropriate.
- XI.E.4 If the Division terminates this general permit, it will provide written notice to affected owner or operators prior to the termination of the general permit. The notice will advise owner or operators that they must apply for a source-specific construction permit as required by Regulation Number 3, Part B.
- XI.E.5 The Division may require any well production facility authorized by a general construction permit to apply for and obtain a source-specific construction permit if circumstances have changed since the time of the original general permit application so that the well production facility is no longer appropriately controlled and/or permitted under the general construction permit. (Regulation Number 3, Part B, Section III.1.3.c.(i)(B))

XII. Additional Requirements

- XII.A Storage tanks registered under this general permit may be subject to 40 CFR, Part 60, Subpart 0000a - Standards of Performance for Crude Oil and Natural Gas Facilities for which Construction, Modification, or Reconstruction Commenced after September 18, 2015. 81 Fed. Reg. 35824 (June 3, 2016). This rule has not yet been incorporated



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

into the Commission's Regulation Number 6. A copy of the complete subpart is available at the Office of the Federal Register website.

- XII.B Storage tanks registered under this general permit may be subject to 40 CFR, Part 60, Subpart 0000b - Standards of Performance for Crude Oil and Natural Gas Facilities for which Construction, Modification, or Reconstruction Commenced after December 6, 2022.89 Fed. Reg. 16820 (Mar. 8, 2024). This rule has not yet been incorporated into Commission's Regulation Number 6. A copy of the complete subpart is available at the [Office of the Federal Register website](#).
- XII.C Natural gas-fired spark ignition engines registered under this general permit may be subject to 40 CFR, Part 60, Subpart JJJJ—Standards of Performance for Stationary Spark Ignition Internal Combustion Engines.73 Fed. Reg. 3568 (Jan. 18, 2008) . This rule has not yet been incorporated into the Commission's Regulation Number 6. A copy of the complete subpart is available on the [EPA's website - Standards of Performance for Stationary Spark Ignition Internal Combustion Engines - Subpart JJJJ](#).
- XII.D Natural gas-fired spark ignition engines registered under this general permit may be subject to 40 CFR, Part 63, Subpart ZZZZ - National Emission Standards for Hazardous Air Pollutants for Reciprocating Internal Combustion Engines. 73 Fed. Reg. 3568 (Jan. 18, 2008). The January 18, 2008 amendments to include requirements for area sources and engines $\leq$ 500 hp located at major sources have not yet been incorporated into the Commission's Regulation Number 8. A copy of the complete subpart is available on the [EPA's website - National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines - Subpart ZZZZ](#). Additional information regarding area source standards can be found on the [EPA guidance website](#).



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

XII.E Natural gas-fired spark ignition engines registered under this general permit may be subject to 40 CFR, Part 63, Subpart ZZZZ - National Emission Standards for Hazardous Air Pollutants for Reciprocating Internal Combustion Engines. 75 Fed. Reg. 51570 (Aug. 20, 2010). The August 20, 2010 amendments to include requirements for existing engines located at area sources and existing engines $\leq$ 500 hp located at major sources have not yet been incorporated into the Commission's Regulation Number 8. A copy of the complete subpart is available on the [EPA website - National Emission Standards for Hazardous Air Pollutants - Subpart ZZZZ](#). Additional information regarding area source standards can be found on the [EPA guidance website](#).



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Attachment A: Alternative Operating Scenarios - Natural gas-fired reciprocating internal combustion engines

1. The following AOS for the temporary and permanent replacement of natural gas fired reciprocating internal combustion engines has been reviewed in accordance with the requirements of Regulation Number 3., Part A, Section IV.A, Operational Flexibility-Alternative Operating Scenarios, Regulation Number 3, Part B, Construction Permits, and Regulation Number 3, Part D, Major Stationary Source New Source Review and Prevention of Significant Deterioration, and it has been found to meet all applicable substantive and procedural requirements. This general permit incorporates and is considered to be a Construction Permit for any engine replacement performed in accordance with this AOS, and the owner or operator is allowed to perform such engine replacement without applying for a revision to this general permit or obtaining a new Construction Permit.
2. The following AOS is incorporated into this general permit in order to deal with an engine breakdown or periodic routine maintenance and repair of an existing onsite engine that requires the use of either a temporary or permanent replacement engine. The compliance demonstrations and any periodic monitoring required by this AOS are in addition to any compliance demonstrations or periodic monitoring required by this general permit.
3. Prohibitions
 - 3.1. The AOS **cannot be used** for the permanent replacement of an entire engine at any source that becomes a major stationary source for purposes of Prevention of Significant Deterioration or Non-Attainment Area New Source Review ("PSD/NANSR") unless the existing engine has emission limits that are below the significance levels in Regulation Number 3, Part D, II.A.44. Nothing in this AOS



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

precludes the Division from taking an action, based on any permanent engine replacement(s), for circumvention of any state or federal PSD/NANSR requirement. Additionally, in the event that any permanent engine replacement(s) constitute(s) a circumvention of applicable PSD/NANSR requirements, nothing in this AOS excuses the owner or operator from complying with PSD/NANSR and applicable permitting requirements.

- 3.2. This AOS **cannot be used** to change the annual permitted emission limits or any compliance emission factor or calculation method that is required by a permit to be used for the purposes of monitoring compliance with the annual permitted emission limits.
- 3.3. This AOS **cannot be used** for the permanent replacement of an engine subject to an emission limitation under Regulation Number 26, Part B, Section I.D.5 (state-wide requirements for natural gas-fired reciprocating internal combustion engines with manufacturer's design rate greater than or equal to 1000 hp), unless the replacement is subject to the same numerical emission standard and compliance date as the unit being replaced, on the date that the replacement will occur.
- 3.4. This AOS cannot be used for the permanent replacement of an engine subject to an emission limitation under Regulation Number 26, Part B, Section I.D.6 (state-wide requirements for rich burn and lean burn natural gas-fired reciprocating internal combustion engines with manufacturer's design rate greater than or equal to 100 hp and 250 hp, respectively, and less than 1000 hp), unless the replacement is subject to the same numerical emission standard and compliance date as the unit being replaced, on the date that the replacement will occur.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

3.5. **Additional Sources:** The replacement of an existing engine with a new engine is viewed by the Division as the installation of a new emissions unit, not “routine replacement” of an existing unit. The AOS is therefore essentially an advanced GP12 registration review. The AOS cannot be used for additional new emission points for any site; an engine that is being installed as an entirely new emission point and not as part of an AOS-approved replacement of an existing onsite engine has to go through the appropriate Construction/Operating permitting process prior to installation.

4. General Provisions

4.1. All replacement engines are subject to all federally applicable and state-only requirements set forth in this general permit for the replaced engine (including monitoring and recordkeeping), unless otherwise superseded by the applicable requirements identified in the Applicability Report for a particular permanently replaced engine.

4.2. The results of all tests and the associated calculations required by this AOS must be submitted to the Division within 30 calendar days of the test or within 60 days of the test if such testing is required to demonstrate compliance with NSPS or MACT requirements. Results of all tests must be maintained and made available to the Division upon request.

4.3. **Contemporaneous Log:** The owner or operator must maintain a log on-site and contemporaneously record the start and stop date of any engine replacement, the manufacturer, date of manufacture, model number, horsepower, and serial number of the engine(s) that are replaced during the term of this general permit, and the manufacturer, model number, horsepower, and serial number of the replacement engine. In addition to the log, for permanent engine



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

replacements, the owner or operator must maintain a copy of all Applicability Reports required under Condition 6 below and make them available to the Division upon request.

4.4. Definition of Temporary Replacement and Permanent Replacement

4.4.1. “Temporary” is defined as in the same service for either 90 operating days or 270 operating days or less in any 12 month period, based on the total number of days that the engine is in operation. If the engine operates only part of a day, that day counts as a full day towards the total. Except as provided below for natural gas-fired reciprocating internal combustion engines subject to an emission limitation under Regulation Number 26, Part B, Section I.D.5 or I.D.6., temporary is defined as 90 operating days for engines where permanent replacements are allowed or 270 operating days where permanent replacements are not allowed.

4.4.2. “Temporary” for natural gas-fired reciprocating internal combustion engines subject to an emission limitation under Regulation Number 26, Part B, Section I.D.5 or I.D.6. is defined as 270 days operating days.

4.4.3. “Permanent” is defined as in the same service for more operating days than allowed under temporary replacement, as defined for each specific engine.

5. Specific Provisions for Temporary Replacements: The owner or operator may temporarily replace an existing engine that is subject to the emission limits set forth in this general permit with an engine that is of the same manufacturer, model, and horsepower or a different manufacturer, model, or horsepower as the existing engine without modifying the GP12 registration, so long as the temporary replacement



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

engine complies with all permit limitations and other requirements applicable to the existing engine. The owner or operator must also comply with any additional requirements applicable to the temporary replacement. Measurement of emissions from the temporary replacement engine must be made as set forth in Condition 8 below.

6. Specific Provisions for Permanent Replacements

- 6.1. Except as provided in Condition 5, the owner or operator may permanently replace an existing engine with another engine with the same manufacturer, model, and horsepower without modifying the GP12 registration so long as the permanent replacement engine complies with all permit limitations and requirements applicable to the existing engine, unless otherwise superseded by the regulations identified in the Applicability Report, as well as any other applicable requirements for the replacement engine. Measurement of emissions from the permanent replacement engine and compliance with the applicable emission limitations must be made as set forth in Condition 8 below.
- 6.2. An Air Pollutant Emissions Notice (APEN) that includes the specific manufacturer, model and serial number and horsepower of the permanent replacement engine must be filed with the Division for the permanent replacement engine within 14 calendar days of commencing operation of the replacement engine. The annual emission limitations and associated emission factors requested on the APEN must exactly match the annual emission limitations and emission factors associated with the annual emission limits set forth in the GP12 registration for the existing engine. Any changes to these values cannot be processed as a permanent replacement under this AOS and will instead require a GP12 registration modification.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 6.3. The APEN must be accompanied by the appropriate APEN filing fee, a cover letter explaining that the owner or operator is exercising an alternative operating scenario and is installing a permanent replacement engine, and a copy of the relevant Applicability Reports for the replacement engine. Applicability Reports can be found at [Divisions' alternate operating scenarios webpage](#). This submittal must be accompanied by a certification from the person legally authorized to act on behalf of the owner or operator indicating that “based on the information and belief formed after reasonable inquiry, the statements and information included in the submittal are true, accurate and complete”.
- 6.4. Annual emission calculations for monitoring compliance with annual permit limits for permanent replacement engines must be performed using the emission factors and calculation methods set forth in the GP12 registration for the existing engine. If the owner or operator elects to modify either the annual emission limit or the emission factor in the GP12 registration associated with annual limit to more closely reflect the replacement engine operation, a GP12 registration modification must be submitted to the Division. The use of any emission factors or calculation methods for the purposes of monitoring compliance with annual limits is prohibited until a complete application to modify the GP12 registration is received by the Division.
- 6.5. Nothing in this AOS is intended to require the owner or operator to use emission factors associated with annual permit limits to monitor compliance with any other emission limits or standards, unless the owner or operator is already explicitly required to do so by a specific permit condition or applicable requirement.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 6.6. The owner or operator must pay fees based on the normal permit processing rate for review of information submitted to the Division in regard to any permanent engine replacement.
7. Applicable Regulations for Permanent Engine Replacements: An analysis of all applicable requirements, including but not limited to monitoring, recordkeeping and reporting, for the permanent engine replacement must be included in the Applicability Reports required under Condition 6. Any testing required by the rules listed in Conditions 7.2 through 7.10 below is in addition to that required by this AOS.
 - 7.1. Reasonably Available Control Technology (RACT) - Regulation Number 3, Part B, Section III.D.2.: All permanent replacement engines that are located in an area that is classified as attainment/maintenance or nonattainment must apply Reasonably Available Control Technology (RACT) for the pollutants for which the area is attainment/maintenance or nonattainment. Note that both VOC and NOx are precursors for ozone. RACT must be applied for any level of emissions of the pollutant for which the area is in attainment/maintenance or nonattainment, except as follows:
 - 7.1.1. In the Denver Metropolitan PM10 attainment/maintenance area, RACT applies to PM10 at any level of emissions and to NOx and SO2, as precursors to PM10, if the potential to emit of NOx or SO2 exceeds 40 tons/yr.
 - 7.1.2. For purposes of this AOS, the following are considered RACT for natural-gas fired reciprocating internal combustion engines:
 - 7.1.2.1. VOC: The emission limitations in NSPS JJJJ.
 - 7.1.2.2. CO: The emission limitations in NSPS JJJJ.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

7.1.2.3. NO_x: The emission limitations in NSPS JJJJ.

7.1.2.4. SO₂: Use of natural gas as fuel.

7.1.2.5. PM₁₀: Use of natural gas as fuel.

7.1.3. As defined in 40 CFR Part 60 Subparts GG (§ 60.331) and 40 CFR Part 72 (§ 72.2), natural gas contains 20.0 grains or less of total sulfur per 100 standard cubic feet.

7.2. Regulation Number 26, Part B, Section I.A - Requirements for new and existing engines (in the 8-hour Ozone Control Area). 8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1.

7.2.1. Natural gas-fired stationary source in the 8-hour Ozone Control Area with manufacturer's design rate greater than 500 horsepower shall employ a non-selective catalyst reduction system and air fuel controller (for rich burn engines), or an oxidation catalyst (for lean burn engines).

7.2.2. For specific rule requirements, see:

7.2.2.1. Section I.A - Applicability and compliance dates

7.2.2.2. Section I.B - Requirements for Control Devices

7.2.2.3. Section I.C - Exemptions

7.3. Regulation Number 26, Part B, Section I.D.3 - New, Modified and Relocated Natural Gas Fired Reciprocating Internal Combustion Engines (State Only)

7.3.1. A natural gas-fired reciprocating internal combustion engines that is either constructed or relocated to the state of Colorado from another



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

state after the date listed in the table are is subject to the emission standards in the *Table 8 - Table 1 from Regulation Number 26, Part B*.

Table 8 - Table 1 from Regulation Number 26, Part B

Max Engine (HP)	Construction or Relocation Date	NO _x (g/hp-hr)	CO (g/hp-hr)	VOC (g/hp-hr)
100≤hp<500	January 1, 2008	2.0	4.0	1.0
	January 1, 2011	1.0	2.0	0.7
500≤hp	July 1, 2007	2.0	4.0	1.0
	July 1, 2010	1.0	2.0	0.7

7.3.2. For specific rule requirements, see:

7.3.2.1. Section I.D.1 - Exemptions.

7.3.2.2. Section I.D.2 - General Provisions

7.3.2.3. XIV.G.3.b. (ii) Section I.D.3 - Emission Standards.

7.4. Regulation Number 26, Part B, Section I.D.4 - Existing Natural Gas Fired Reciprocating Internal Combustion Engines



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 7.4.1. Rich burn natural gas-fired reciprocating internal combustion engines with manufacturer's name plate design rate greater than 500 horsepower constructed or modified before February 1, 2009 shall install and operate a non-selective catalyst reduction system and air fuel controller.
- 7.4.2. (State-Only) All lean burn natural gas-fired reciprocating internal combustion engines with manufacturer's name plate design rate greater than 500 horsepower shall install and operate an oxidation catalyst.
- 7.4.3. For specific rule requirements, see:
 - 7.4.3.1. Section I.D.1 - Exemptions
 - 7.4.3.2. Section I.D.4.a.(i)(B)-(C) - Exemptions
 - 7.4.3.3. Section I.D.4.a.(i)(A) - General Provisions
 - 7.4.3.4. Section I.D.4.a.(i) - Control Requirements
 - 7.4.3.5. Section I.D.4.b.(i) - Control Requirements
 - 7.4.3.6. Section I.D.4.c. - Control Requirements listed engines ≥ 1000 HP
- 7.5. Additional Requirements for Natural Gas Fired Reciprocating Internal Combustion Engines: Regulation Number 26, Part B, Section I.D.5 (State-Only).
 - 7.5.1. Stationary natural gas-fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 1000 horsepower are subject to specific emission standards. The specific applicable standards and compliance date of any particular engine or



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

any particular replacement of that engine at any particular point in time is variable, dependent on multiple factors, and can potentially be subject to further changes at the discretion of the owner or operator.

7.5.2. For specific rule requirements, see:

7.5.2.1. Section I.D.5.a. - Applicability and exemptions

7.5.2.2. Section I.D.5.b. - Standards for engines not under a company-wide plan

7.5.2.3. Section I.D.5.c. - Alternative Company-Wide Compliance Plan

7.5.2.4. Section I.D.5.d. - Performance Testing

7.5.2.5. Section I.D.5.e. - Monitoring and Combustion Process Adjustment

7.5.2.6. Section I.D.5.f. - Recordkeeping

7.5.2.7. Section I.D.5.g - Reporting.

7.6. Additional Requirements for Natural Gas Fired Reciprocating Internal Combustion Engines: Regulation Number 26, Part B, Section I.D.6 (State-Only).

7.6.1. Rich burn stationary natural gas-fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 100 horsepower but less than 1000 horsepower and lean burn stationary natural gas-fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 250 horsepower, but less than 1000 horsepower are subject to specific emission standards. The specific applicable standards and compliance date of



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

any particular engine or any particular replacement of that engine at any particular point in time is variable, dependent on multiple factors, and can potentially be subject to further changes at the discretion of the owner or operator.

7.6.2. For specific rule requirements, see:

7.6.2.1. Section I.D.6.a - Applicability and exemptions

7.6.2.2. Section I.D.6.b - Emission Standards and Compliance deadlines

7.6.2.3. Section I.D.6.c - Performance Testing

7.6.2.4. Section I.D.6.d - Monitoring and Combustion Process
Adjustment

7.6.2.5. Section I.D.6.e - Recordkeeping

7.6.2.6. Section I.D.6.f - Reporting.

7.7. Control of Emissions from Stationary and Portable Combustion Equipment in the 8-hour Ozone Control Area or Northern Weld County: Regulation Number 26, Part B, Section II. 8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. Northern Weld County is as defined in Regulation Number 7, Part A, Section II.A.3.

7.7.1. Replacements stationary internal combustion engines that were located at a major sources of NO_x prior to specific dates must complete inspections and adjustments.

7.7.2. For specific rule requirements, see:



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

7.7.2.1. Section II.A.1 - Applicability

7.7.2.2. Section II.A.2 - Exemptions

7.7.2.3. Section II.A.6.b.(iv) & (vii) - Combustion Process Adjustment
for Engines

7.7.2.4. Section II.A.6.b.(viii) - Compliance dates and frequency

7.7.2.5. Section II.A.6.c - Alternatives

7.7.2.6. Section II.A.7 - Recordkeeping

7.8. Control of Emissions from Specific Major Sources of VOC and/or NO_x in the 8-hour Ozone Control Area: Regulation Number 26, Part B, Section III. 8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. Replacements of units specifically identified in Section III are subject to the additional requirements of the original unit, as specified in Section III.

7.9. NSPS for stationary spark ignition internal combustion engines: 40 CFR Part 60, Subpart JJJJ

7.9.1. A permanent replacement engine that is manufactured on or after 1/1/2009 for emergency engines greater than 25 hp, 7/1/2008 for engines less than 500 hp, 7/1/2007 for engines greater than or equal to 500 hp except for lean burn engines greater than or equal to 500 hp and less than 1,350 hp, and 1/1/2008 for lean burn engines greater than or equal to 500 hp and less than 1,350 hp are subject to the requirements of 40 CFR Part 60, Subpart JJJJ.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 7.9.2. Note that the initial test required by NSPS Subpart JJJJ can serve as the testing required by this AOS under Condition 8, if approved in advance by the Division, provided that such test is conducted within the time frame specified in Condition 8.
- 7.9.3. Under the provisions of Regulation Number 6, Part B, Section I.C, which are triggered upon the adoption of NSPS JJJJ, relocation of an internal combustion engine (ICE) is defined as the earlier of the date the owner or operator submits an APEN to the Division, or the date the ICE is placed in a secured location where it is intended to be operated within the State of Colorado. However, as of March 23, 2023 the Division has not yet adopted NSPS JJJJ. Until such time as it does, any engine subject to NSPS JJJJ will be subject only under Federal law. Once the Division adopts NSPS JJJJ, determination of the applicable State-Only Enforceable requirements of NSPS JJJJ for the purposes of Regulation Number 6 must be based on the relocation date (if any) as defined in § I.C.1.a. Note that special provisions apply to engines < 500 hp. Refer to PS-Memo 10-03 for additional information on Regulation Number 6 applicability for new and relocated engines.
- 7.10. Reciprocating internal combustion engine (RICE) MACT: 40 CFR Part 63, Subpart ZZZZ
- 7.10.1. A permanent replacement engine located at either an area or major source is subject to the requirements in 40 CFR Part 63, Subpart ZZZZ.
- 7.10.2. Note that the initial test required by the MACT can serve as the testing required by this AOS under Condition 8, if approved in advance by the



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Division, provided that such test is conducted within the time frame specified in Condition 8.

8. Portable Analyzer Monitoring

- 8.1. Note: In some cases, there may be conflicting and/or duplicative testing requirements due to overlapping Applicable Requirements. In those instances, please contact the Division's Compliance and Enforcement Unit to discuss streamlining the testing requirements.
- 8.2. Note that the testing required by this Condition may be used to satisfy the portable analyzer monitoring requirements specified by the permit for the relevant time period (i.e. if the permit requires quarterly portable analyzer monitoring, this monitoring event conducted under the AOS will serve as the quarterly monitoring event required by the permit and an additional portable analyzer monitoring event is not required for another three months).
- 8.3. The owner or operator may conduct a reference method test, in lieu of the portable analyzer test required by this Condition, if approved in advance by the Division.
- 8.4. The owner or operator must measure nitrogen oxide (NO_x) and carbon monoxide (CO) emissions in the exhaust from the replacement engine using a portable flue gas analyzer within seven (7) calendar days of commencing operation of the replacement engine.
- 8.5. All portable analyzer monitoring required by this general permit must be conducted using the Division's Portable Analyzer Monitoring Protocol (version March 2006 or newer) as found on the Division's web site at:
<https://cdphe.colorado.gov/compliance-and-enforcement>



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 8.6. Results of the portable analyzer monitoring must be used to monitor the compliance status of this unit, except for any specifically applicable emission limitations or standards for which an applicable requirement specifies a performance test or Continuous Emission Monitoring System as the compliance monitoring method.
- 8.7. For comparison with an annual (tons/year) or short term (lbs/unit of time) emission limit, the results of the monitoring event must be converted to a lb/hr basis and multiplied by the allowable operating hours in the month or year (whichever applies) in order to monitor compliance. If a source is not limited in its hours of operation the test results will be multiplied by the maximum number of hours in the month (744) or year (8760), whichever applies.
- 8.8. For comparison with a short-term limit that is either input based (lb/mmBtu), output based (g/hp-hr) or concentration based (ppmvd @ 15% O₂) that the existing unit is currently subject to or the replacement engine will be subject to, the results of the test must be converted to the appropriate units as described in the above-mentioned Portable Analyzer Monitoring Protocol document.
- 8.9. If the portable analyzer results indicate compliance with both the NO_x and CO emission limitations, in the absence of credible evidence to the contrary, the owner or operator may certify that the engine complies with both the NO_x and CO emission limitations for the relevant time period.
- 8.10. Subject to the provisions of § 25-7-123.1, C.R.S. and in the absence of credible evidence to the contrary, if the portable analyzer results fail to demonstrate compliance with either the NO_x or CO emission limitations, the engine will be



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

considered to be out of compliance from the date of the portable analyzer test until a portable analyzer test indicates compliance with both the NO_x and CO emission limitations or until the engine is taken offline.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

Attachment B: Alternative Operating Scenarios - Diesel fuel-fired reciprocating internal combustion engines

1. The following AOS for the temporary and permanent replacement of Stationary (CI) engines has been reviewed in accordance with the requirements of Regulation Number 3, Part A, Section IV.A, Operational Flexibility- Alternative Operating Scenarios; Regulation Number 3, Part B, Construction Permits; and Regulation Number 3, Part D, Major Stationary Source New Source Review and Prevention of Significant Deterioration and has been found to meet all applicable substantive and procedural requirements. This general permit incorporates and shall be considered a Construction Permit for any engine replacement performed in accordance with this AOS, and the owner or operator shall be allowed to perform such engine replacement without applying for a revision to this general permit or obtaining a new Construction Permit. Depending on the type of replacement, the owner or operator may be required to submit an application to modify the GP12 registration.
2. The following AOS is incorporated into this general permit in order to deal with an engine breakdown or periodic routine maintenance and repair of an existing onsite engine that requires the use of either a temporary or permanent replacement engine. The compliance demonstrations and any periodic monitoring required by this AOS are in addition to any compliance demonstrations or periodic monitoring required by this general permit.
3. Prohibitions
 - 3.1. The AOS **cannot be used** for the permanent replacement of an entire engine at any source that becomes a major stationary source for purposes of Prevention of Significant Deterioration or Non-Attainment Area New Source Review ("PSD/NANSR") unless the existing engine has emission limits that are below the



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

significance levels in Regulation Number 3, Part D, II.A.44. Nothing in this AOS precludes the Division from taking an action, based on any permanent engine replacement(s), for circumvention of any state or federal PSD/NANSR requirement. Additionally, in the event that any permanent engine replacement(s) constitute(s) a circumvention of applicable PSD/NANSR requirements, nothing in this AOS excuses the owner or operator from complying with PSD/NANSR and applicable permitting requirements.

- 3.2. This AOS **cannot be used** to change the annual permitted emission limits or any compliance emission factor or calculation method that is required by a permit to be used for the purposes of monitoring compliance with the annual permitted emission limits.
- 3.3. This AOS cannot be used for the permanent replacement of an engine subject to an emission limitation under Regulation Number 26, Part B, Section I.D.6, unless the replacement is subject to the same numerical emission standard and compliance date as the unit being replaced, on the date that the replacement will occur.

4. General Provisions

- 4.1. All replacement engines are subject to all federally applicable and state-only requirements set forth in this general permit for the replaced engine (including monitoring and recordkeeping), unless otherwise superseded by the applicable requirements identified in the Applicability Report for a particular permanently replaced engine.
- 4.2. The results of all tests and the associated calculations required by this AOS must be submitted to the Division within 30 calendar days of the test or within 60 days of the test if such testing is required to demonstrate compliance with



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

NSPS or MACT requirements. Results of all tests must be maintained and made available to the Division upon request.

- 4.3. Contemporaneous Log: The owner or operator must maintain a log on-site and contemporaneously record the start and stop date of any engine replacement, the manufacturer, date of manufacture, model number, horsepower, and serial number of the engine(s) that are replaced during the term of this general permit, and the manufacturer, model number, horsepower, and serial number of the replacement engine. In addition to the log, for permanent engine replacements, the owner or operator must maintain a copy of all Applicability Reports required under Condition 6 below and make them available to the Division upon request.

4.4. Definition of Temporary Replacement and Permanent Replacement

- 4.4.1. "Temporary" is defined as in the same service for either 90 operating days or 270 operating or less in any 12 month period, based on the total number of days that the engine is in operation. If the engine operates only part of a day, that day counts as a full day towards the total. Except as provided below for diesel-fired reciprocating internal combustion engines subject to an emission limitation under Regulation Number 26, Part B, Section I.D.6., temporary is defined as 90 operating days for engines where permanent replacements are allowed or 270 operating days where permanent replacements are not allowed.
- 4.4.2. "Temporary" for diesel-fired reciprocating internal combustion engines subject to an emission limitation under Regulation Number 26, Part B, Section I.D.6. is defined as 270 days operating days.



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 4.4.3. “Permanent” is defined as in the same service for more operating days than allowed under temporary replacement, as defined for each specific engine.
5. Specific Provisions for Temporary Replacements: The owner or operator may temporarily replace an existing engine that is subject to the emission limits set forth in this general permit with an engine that is of the same manufacturer, model, and horsepower or a different manufacturer, model, or horsepower as the existing engine without modifying the GP12 registration, so long as the temporary replacement engine complies with all permit limitations and other requirements applicable to the existing engine. The owner or operator must also comply with any additional requirements applicable to the temporary replacement. Measurement of emissions from the temporary replacement engine must be made as set forth in Condition 8 below.
6. Specific Provisions for Permanent Replacements
- 6.1. Except as provided in Condition 5, the owner or operator may permanently replace an existing engine with another engine with the same manufacturer, model, and horsepower without modifying the GP12 registration so long as the permanent replacement engine complies with all permit limitations and requirements applicable to the existing engine, unless otherwise superseded by the regulations identified in the Applicability Report, as well as any other applicable requirements for the replacement engine. Measurement of emissions from the permanent replacement engine and compliance with the applicable emission limitations must be made as set forth in Condition 8 below.
- 6.2. An Air Pollutant Emissions Notice (APEN) that includes the specific manufacturer, model and serial number and horsepower of the permanent



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

replacement engine must be filed with the Division for the permanent replacement engine within 14 calendar days of commencing operation of the replacement engine. The annual emission limitations and associated emission factors requested on the APEN must exactly match the annual emission limitations and emission factors associated with the annual emission limits set forth in the GP12 registration for the existing engine. Any changes to these values cannot be processed as a permanent replacement under this AOS and will instead require a GP12 registration modification.

- 6.3. The APEN must be accompanied by the appropriate APEN filing fee, a cover letter explaining that the owner or operator is exercising an alternative operating scenario and is installing a permanent replacement engine, and a copy of the relevant Applicability Reports for the replacement engine. Applicability Reports can be found at [Divisions' alternate operating scenarios webpage](#). This submittal must be accompanied by a certification from the person legally authorized to act on behalf of the owner or operator indicating that "based on the information and belief formed after reasonable inquiry, the statements and information included in the submittal are true, accurate and complete".
- 6.4. Annual emission calculations for monitoring compliance with annual permit limits for permanent replacement engines must be performed using the emission factors and calculation methods set forth in the GP12 registration for the existing engine. If the owner or operator elects to modify either the annual emission limit or the emission factor in the GP12 registration associated with annual limit to more closely reflect the replacement engine operation, a GP12 registration modification must be submitted to the Division. The use of any emission factors or calculation methods for the purposes of monitoring



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

compliance with annual limits is prohibited until a complete application to modify the GP12 registration is received by the Division.

- 6.5. Nothing in this AOS is intended to require the owner or operator to use emission factors associated with annual permit limits to monitor compliance with any other emission limits or standards, unless the owner or operator is already explicitly required to do so by a specific permit condition or applicable requirement.
- 6.6. The owner or operator must pay fees based on the normal permit processing rate for review of information submitted to the Division in regard to any permanent engine replacement.
- 6.7. The replacement engine's compliance with the applicable emission limitations of the original engine shall be monitored by one of the following methods:
 - 6.7.1. Manufacturer certified emission factors showing compliance.
 - 6.7.2. Stack tests of same make and model showing compliance. This would only be considered if the test was done under similar conditions to Colorado (i.e. at altitude).
 - 6.7.3. Stack tests on the engine.
7. Applicable Regulations for Permanent Engine Replacements
 - 7.1. NSPS for stationary compression ignition internal combustion engines: 40 CFR Part 60, Subpart IIII.
 - 7.1.1. A permanent replacement engine that is ordered after July 11, 2005 and manufactured after April 1, 2006 or is modified or reconstructed



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

after July 11, 2005 is subject to the requirements of 40 CFR Part 60, Subpart IIII. An analysis of applicable monitoring, recordkeeping, and reporting requirements for the permanent engine replacement shall be included in any request for a permanent engine replacement.

- 7.1.2. Note that under the provisions of Regulation Number 6, Part B, Section I.B. that relocation of a source from outside of the State of Colorado into the State of Colorado is considered to be a new source, subject to the requirements of Regulation Number 6 (i.e., the date that the source is first relocated to Colorado becomes equivalent to the date of manufacture for purposes of determining the applicability of NSPS IIII requirements).
- 7.2. MACT for Stationary Reciprocating Internal Combustion Engines: 40 CFR Part 63, Subpart ZZZZ.
 - 7.2.1. Any permanent replacement engine located at either an area or major source is subject to the requirements of 40 CFR Part 63, Subpart ZZZZ. An analysis of applicable monitoring, recordkeeping, and reporting requirements for the permanent engine replacement shall be included in any request for a permanent engine replacement.
- 7.3. Additional Requirements for Natural Gas Fired Reciprocating Internal Combustion Engines: Regulation Number 26, Part B, Section I.D.6 (State-Only).
 - 7.3.1. stationary diesel or dual-fuel fired internal combustion engines state-wide with a manufacturer's design rate greater than or equal to 500 horsepower. The specific applicable standards and compliance date of any particular engine or any particular replacement of that engine at any particular point in time is variable, dependent on multiple factors,



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

and can potentially be subject to further changes at the discretion of the owner or operator.

7.3.2. For specific rule requirements, see:

7.3.2.1. Section I.D.6.a - Applicability and exemptions

7.3.2.2. Section I.D.6.b - Emission Standards and Compliance deadlines

7.3.2.3. Section I.D.6.c - Performance Testing

7.3.2.4. Section I.D.6.d - Monitoring and Combustion Process
Adjustment

7.3.2.5. Section I.D.6.e - Recordkeeping

7.3.2.6. Section I.D.6.f - Reporting.

8. Portable Analyzer Monitoring

8.1. Note: In some cases, there may be conflicting and/or duplicative testing requirements due to overlapping Applicable Requirements. In those instances, please contact the Division's Compliance and Enforcement Unit to discuss streamlining the testing requirements.

8.2. Note that the testing required by this Condition may be used to satisfy the portable analyzer monitoring requirements specified by the permit for the relevant time period (i.e. if the permit requires quarterly portable analyzer monitoring, this monitoring event conducted under the AOS will serve as the quarterly monitoring event required by the permit and an additional portable analyzer monitoring event is not required for another three months).



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

- 8.3. The owner or operator may conduct a reference method test, in lieu of the portable analyzer test required by this Condition, if approved in advance by the Division.
- 8.4. The owner or operator must measure nitrogen oxide (NO_x) and carbon monoxide (CO) emissions in the exhaust from the replacement engine using a portable flue gas analyzer within seven (7) calendar days of commencing operation of the replacement engine.
- 8.5. All portable analyzer monitoring required by this general permit must be conducted using the Division's Portable Analyzer Monitoring Protocol (ver March 2006 or newer) as found on the Division's web site at:
<https://cdphe.colorado.gov/compliance-and-enforcement>
- 8.6. Results of the portable analyzer monitoring must be used to monitor the compliance status of this unit, except for any specifically applicable emission limitations or standards for which an applicable requirement specifies a performance test or Continuous Emission Monitoring System as the compliance monitoring method.
- 8.7. For comparison with an annual (tons/year) or short term (lbs/unit of time) emission limit, the results of the monitoring event must be converted to a lb/hr basis and multiplied by the allowable operating hours in the month or year (whichever applies) in order to monitor compliance. If a source is not limited in its hours of operation the test results will be multiplied by the maximum number of hours in the month (744) or year (8760), whichever applies.
- 8.8. For comparison with a short-term limit that is either input based (lb/mmBtu), output based (g/hp-hr) or concentration based (ppmvd @ 15% O₂) that the



COLORADO

Air Pollution Control Division

Department of Public Health & Environment

existing unit is currently subject to or the replacement engine will be subject to, the results of the test must be converted to the appropriate units as described in the above-mentioned Portable Analyzer Monitoring Protocol document.

- 8.9. If the portable analyzer results indicate compliance with both the NO_x and CO emission limitations, in the absence of credible evidence to the contrary, the owner or operator may certify that the engine complies with both the NO_x and CO emission limitations for the relevant time period.
- 8.10. Subject to the provisions of § 25-7-123.1, C.R.S. and in the absence of credible evidence to the contrary, if the portable analyzer results fail to demonstrate compliance with either the NO_x or CO emission limitations, the engine will be considered to be out of compliance from the date of the portable analyzer test until a portable analyzer test indicates compliance with both the NO_x and CO emission limitations or until the engine is taken offline.